

RFP for "Supply, Installation, Commissioning, Integration, and Setup of HCI Servers and Virtualization Infra at MP Tender No. – MPSEDC/SDC/2026/669

Query Responses

Note: The responses mentioned in this sheet is only for clarification purposes to the bidders. Though utmost care has been taken to provide accurate information, any ambiguity in responses should be clarified by the bidder and RFP and Corrigendum Clauses and amendments shall prevail.

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
1	13	2. Eligibility/Qualification Criteria Point No. 03	The bidder must have successfully completed assignment of similar nature involving the Supply, Installation, Configuring, and Maintenance of Data Centre or Disaster Recovery site projects of IT infra implementation and O&M within India Provided that project shall not be of the period prior to 1st April 2020 and must have undertaken for any of the following: A) State or Central Government OR B) State or Central PSU OR C) Scheduled Commercial Banks (as per RBI's latest notification) The below mentioned order values are applicable for the above criteria: The bidder should have undertaken One project of a similar nature with a minimum value of INR 14 Crores. OR Two projects of a similar nature, each with a minimum value of INR 10 Crores. OR Three projects of a similar nature, each with a minimum value of INR 07 Crores Note: Only supply orders shall not be considered. The Bidder should have provided FMS/Data Centre Operation for such projects.	We request you to kindly consider the work order effective from 1 January 2018 instead of 1 April 2020. The bidder must have experience in assignment of similar nature involving the Supply, Installation, Configuring, and Maintenance of Data Centre or Disaster Recovery site projects of IT infra implementation and O&M within India Provided that project shall not be of the period prior to 1st Jan 2018 and must have undertaken for any of the following: A) State or Central Government OR B) State or Central PSU OR C) Scheduled Commercial Banks (as per RBI's latest notification) The below mentioned order values are applicable for the above criteria: The bidder should have undertaken One project of a similar nature with a minimum value of INR 14 Crores. OR Two projects of a similar nature, each with a minimum value of INR 10 Crores. OR Three projects of a similar nature, each with a minimum value of INR 07 Crores Note: Only supply orders shall not be considered. The Bidder should have provided FMS/Data Centre Operation for such projects.	Refer Corrigendum-2 Sn#12
2	24	5. Project Timeline & Payment Milestone:	Supply of complete Infrastructure as per Financial Proposal: Project Timeline: (T+ 6) week Payment Milestone: 20% of the Work Order Value	The Standard delivery lead time for Server & Networking components is 12-14 Weeks. Project Timeline: (T+ 12-14) week Payment Milestone: 40% of the Work Order Value	Refer Corrigendum-2 Sn#1

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3	24	5. Project Timeline & Payment Milestone:	Installation, commissioning of all Infrastructure & system UAT / Verification Project Timeline: (T+ 8) week Payment Milestone: 20% of the Work Order Value	Project Timeline: (T+ 14-16) week Payment Milestone: 30% of the Work Order Value	Refer Corrigendum-2 Sn#2
4	24	5. Project Timeline & Payment Milestone:	Integration with Existing infra/ solution and FAT of all the Installed, commissioned Infrastructure and Go-Live (system is in usable state). Project Timeline: (T+ 10) week Payment Milestone: 50% of the Work Order Value	Project Timeline: (T+ 16-18) week Payment Milestone: 20% of the Work Order Value	Refer Corrigendum-2 Sn#3
5	25	6.1. SLAs for Supply, Installation and Commissioning of Infrastructure:	Supply of complete Infrastructure (as per BOQ) Timeline: (T + 6) week	Timeline: (T + 12-14) week	Refer Corrigendum-2 Sn#4
6	25	6.1. SLAs for Supply, Installation and Commissioning of Infrastructure:	Installation, Commissioning, system ready for UAT Timeline: (T + 8) week	Timeline: (T + 14-16) week	Refer Corrigendum-2 Sn#4
7	25	6.1. SLAs for Supply, Installation and Commissioning of Infrastructure:	FAT of all the supplied components Timeline: (T + 10) week	Timeline: (T + 16-18) week	Refer Corrigendum-2 Sn#4
8	Page no 16	Scope of work 3.1 Solution Design & Architecture	Resiliency: Data resiliency with FTT- 1/RF-2, with future support for FTT-2/RF-3.	Modification : Resiliency: Data resiliency with FTT- 2/RF-3 to support SLA & availability of Critical SDC workload , Solution must be offered with 1Ms response time with asked IOPS of 100000 70:30 R:W- 16KB Block size Justification: Resiliency for Critical SDC workload, FTT2/RF3 will provide 2 Node failover resiliency , This will ensure availability required for SDC critical workload	Refer Corrigendum-2 Sn#5

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
9	Page no 20	4. Technical Specification and Compliance Point#1	Offered Solution shall be a next generation software defined platform which shall offer the functionality of Hyper-converged with independent scaling of both Storage and Compute without any downtime. The solution should provide enterprise-class storage services using latest x86 server infrastructures without dependence on a separate Storage Area Network Storage & associated components such as SAN Switches & HBAs. Offered HCI system should be a software-centric system for delivering compute, storage and networking resources in a tightly integrated system	Modification Request: Offered Solution shall be a next generation software defined platform which shall offer the functionality of Hyper-converged with independent scaling of both Storage and Compute without any downtime. The solution should provide enterprise-class storage services using latest x86 server infrastructures without dependence on a separate Storage Area Network Storage & associated components such as SAN Switches & HBAs. Offered HCI system should be a software-centric system for delivering compute, storage and networking resources in a tightly integrated HCI solution to deliver asked HCI services. Justification: In HCI solution offering, different OEM have different architecture to deliver Virtual machine with Performance Integrated with compute, storage and networking resources.	Refer Corrigendum-2 Sn#13
10	Page no 21	4. Technical Specification and Compliance Point#4	Resource requirement: HCI Solution should be provided with N+1 configuration having maximum 12 nodes Hybrid cluster, each node providing latest Intel processors/ AMD processors with base frequency of 2.1GHz or higher and DDR5 Memory. Solution must be proposed with at least 1500 or higher cores with minimum 1300 usable cores, 6 TB or higher memory and with minimum 500 TB usable storage without considering any deduplication, compression, erasure coding or any saving techniques benefits. Complete solution must be proposed with RF2/FTT1 i.e. two copies of data across the cluster. Each server/node should be configured with minimum of one Quad Port 10/25G network ports along having required 25G SFPS/DAC Cables for uplink connectivity	Resource requirement: HCI Solution should be provided with N+1 configuration having maximum 12 nodes Hybrid cluster, each node providing latest Intel processors/ AMD processors with base frequency of 2.1GHz or higher and DDR5 Memory. Solution must be proposed with at least 1500 or higher cores with minimum 1300 usable cores, 6 TB or higher memory and with minimum 500 TB usable storage without considering any deduplication, compression, erasure coding or any saving techniques benefits. Complete solution must be proposed with RF3/FTT2 i.e. two copies of data across the cluster. Each server/node should be configured with minimum of one Quad Port 10/25G network ports along having required 25G SFPS/DAC Cables for uplink connectivity Justification: Resiliency for Critical SDC workload & to ensure SLA with minimum 2 Node Failover architecture	Refer Corrigendum-2 Sn#6

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
11	Page no 21	4. Technical Specification and Compliance Point#2	The proposed HCI solution should be a factory shipped engineered & integrated appliance. The HCI solution should support SSD/NLSAS Drives. The HCI solution should support scalability up to 32 nodes in a single cluster. Each server node should have dedicated redundant hot swap power supply, cooling fans. There should not be any separate disaggregated components in the proposed HCI solution i.e. The 3 Major components of the HCI node i.e. Compute, RAM and Storage should be physically inside the node, no resource should be provisioned from outside the HCI Node.	Modification: The proposed HCI solution should be a factory shipped engineered & integrated appliance. The HCI solution should be offered with NVMe Drives only. The HCI solution should support scalability up to 16 nodes in a single cluster. Each server node should have dedicated redundant hot swap power supply, cooling fans. There should not be any separate disaggregated components in the proposed HCI solution i.e. The 3 Major components of the HCI node i.e. Compute, RAM and Storage should be physically inside the node, no resource should be provisioned from outside the HCI Node or as per OEM architecture ensuring full software defined architecture delivering required performance & IOPS. Justification: 1. Please Change asked Drives to NVMe Drives only , as NL SAS/ SSD /SAS have no Future road Map , Considering this is SDC Critical Environmnet any componnet which is going to be obsolute in near future should not be considered . 2. Change asked architecture with NVMe Drives only , This will ensure performance, abalability & high resileancy with Latest NVMe architectire. 2. Request Change in asked Cluster Scalability as 16/32 Nodes , since 32 Node in single cluster as SDS /HCI solution is not recommended from Management & for future Upgrade & Updated point of view.	Refer Corrigendum-2 Sn#7
12	Page no 21	4. Technical Specification and Compliance Point#5	The HCI solution shall support scaling Compute and Storage capacity and performance linearly by addition of nodes. VMs on existing nodes shall get the storage performance & capacity that gets enhanced by the addition of new HCI nodes in which HCI solution must provide on the fly change of ESE (Enterprise Storage Efficiency)-Deduplication/Compression for workload without any visible impact on storage and their operations and the bidder must ensure minimum 100,000 IOPS from the proposed HCI solution, assuming 8kb block size, and 70:30 read: write ratio.	Modification: The HCI solution shall support scaling Compute and Storage capacity and performance linearly by addition of nodes. VMs on existing nodes shall get the storage performance & capacity that gets enhanced by the addition of new HCI nodes in which HCI solution must provide on the fly change of ESE (Enterprise Storage Efficiency)-Deduplication/Compression for workload without any visible impact on storage and their operations and the bidder must ensure minimum 100,000 IOPS from the proposed HCI solution, assuming 8kb block size, and 70:30 read: write ratio ensuring less than 1ms response time . Justification: Since SDC application & DB Requirement is Critical from Performance POV. Pls define Response time to ensure Performance delivery & architecture is offered	No Change

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
13	Page no 21	4. Technical Specification and Compliance Point#6	The HCI software should pool all HDDs from all the nodes in the cluster to present a single storage resource pool to all server nodes in the cluster. The HCI storage should be a scale-out distributed storage. Proposed HCI solution must have a primary copy of the VM data locally to the Server/Node to offer faster read and write and secondary copy must be distributed on the other HCI Nodes. The intelligent software should consolidate the resources from each server node into a shared resource pool, delivering a high degree of flexibility while also simplifying management.	Modification: The HCI software should pool all NVMe HDDs from all the nodes in the cluster to present a single storage resource pool to all server nodes in the cluster. The HCI storage should be a scale-out distributed storage. Proposed HCI solution must have a primary copy of the VM data locally to the Server/Node to offer faster read and write and secondary copy must be distributed on the other HCI Nodes or as per OEM architecture ensuring less than 1ms sustainable response time. The intelligent software should consolidate the resources from each server node into a shared resource pool, delivering a high degree of flexibility while also simplifying management. Justification: Please change asked Disk type of Storage System to : NVMe. Kindly modify this clause for participation and better protection of investment.	No Change
14	Page no 21	4. Technical Specification and Compliance Point#7	The solution should automatically rebalance data to maintain balanced utilization of storage across the HCI nodes. When storage capacity is scaled up or scaled out, the HCI nodes must automatically redistribute data equally across all nodes equally without migrating VMs. Hypervisor software must provide Data at rest encryption with Native KMS which protects unauthorized data access.	Modification: The solution should automatically rebalance data to maintain balanced utilization of storage across the HCI nodes. When storage capacity is scaled up or scaled out, the HCI nodes must automatically redistribute data equally across all nodes equally without migrating VMs. Hypervisor / HCI solution must provide Data at rest encryption with Native KMS which protects unauthorized data access. Justification: Please change asked Hypervisor Based encryption to HCI Solution based Data at rest encryption - Option for Hardware based Encryption, as per OEM architecture to deliver Data at rest encryption feature	Refer Corrigendum-2 Sn#8
15	Page no 21	4. Technical Specification and Compliance Point#9	HCI solution should support Block, File and Object (S3) natively or using third party solution from Day1. Solution should support file storage supporting NFS v3/v4 and SMB 2.0/3.0 for Linux and Windows Guest with unlimited shares integrated with with Active directory/LDAP	This is Restrictive Clauses, Please Remove this Clause / Modify : HCI solution should support Block, File natively or using third party solution from Day1. Solution should support file storage supporting NFS v3/v4/SMB 2.0/3.0 for Linux and Windows Guest with unlimited shares integrated with with Active directory/LDAP Justification: This is Restrictive Clauses, This point is OEM specific to one OEM Request you to remove the ask of S3 protocol for wider participation of OEM	Refer Corrigendum-2 Sn#17

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
16	Page no 21	4. Technical Specification and Compliance Point#10	The solution quoted shall provide VM level/ Native Storage replication from a Primary site for planned DR Site strategy and vice-versa. If required, bidder can leverage existing replication tool.	The solution quoted shall provide VM level/ Native Storage replication from a Primary site for planned and running DC & DR Site strategy and vice-versa. The proposed solution should have a capability to integrate with existing deployed cloud platform running Visualized and physical hardware. If required, bidder can leverage existing replication tool.	Refer Corrigendum-2 Sn#9
17	Page no 21	4. Technical Specification and Compliance Point#11	Solution should support Virtual Network visibility with application-centric protection from network threats and automation of common networking operations	Modification: Solution should support Virtual Network visibility with application-centric protection from network threats with automation of common existing network operations, security appliances & firewall, load balancer, backup solution. Justification: for seamless integration with existing network, load balancer and security devices. please include the ask modification..	Refer Corrigendum-2 Sn#10
18	Page no 22	4. Technical Specification and Compliance Management Point#2	The management tool should be able to provide global resource pooling and policy management to enable policy-based automation. The proposed management solution should provide proactive security & software advisory alerts and should outline the fixes required to address the issues and the management software should provide a mobile app for monitoring & viewing the resources being managed in the infrastructure	Modification: The management tool should be able to provide global resource pooling and policy management to enable policy-based automation. The proposed management solution should provide proactive security & software advisory alerts and should outline the fixes required to address the issues . Justification: "Security Breach" Please Remove the ask of Mobile app for monitoring & viewing the resources being managed in the infrastructure, as allowing Mobile app for monitoring & viewing the resources will be point of security breach in environment & will have security impact in SDC environment "	Refer Corrigendum-2 Sn#11
19	Page no 22	4. Technical Specification and Compliance Network Switch Point#3	Switch should support minimum 1000 VRF instances with route leaking functionality, minimum 400K IPv4 LPM routes, min 16K Multicast routes, minimum 90K no. of MAC addresses, minimum 32 MB Packet Buffer with intelligent shared-memory buffered architecture to simplify the system buffer management and queuing implementation.	Please amend the clause as: Switch should support minimum 256 VRF instances with route leaking functionality, minimum 120K IPv4 LPM routes, min 7K Multicast routes, minimum 90K no. of MAC addresses, minimum 32 MB Packet Buffer with intelligent shared-memory buffered architecture to simplify the system buffer management and queuing implementation. Justification: 256 VRF are more than sufficient and higher VRF required in ISP environment which supports multiple tenant. 7K multicast routes more than sufficient for current deployment scenario, not more than 100 Multicast will be used.	Refer Corrigendum-2 Sn#14
20	Page no 23	4. Technical Specification and Compliance Network Switch Point#4	Switch should support VXLAN to achieve Network Virtualisation, along with VXLAN and EVPN symmetric IRB. The switch should support BGP EVPN Route Type 2, Type 4 and Route Type 5 for the overlay control plane	Please amend the clause: Switch should support VXLAN to achieve Network Virtualisation, along with VXLAN and EVPN symmetric IRB. The switch should support BGP EVPN Route Type 2, Type 3/Type4 and Route Type 5 for the overlay control plane Justification: request to amend the clause for wider participation.	Refer Corrigendum-2 Sn#15

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
21	Page no 23	4. Technical Specification and Compliance Network Switch Point#8	All ports on Switch must provide encryption of traffic ie. MAC Sec Encryption (802.1AE) in hardware and should support QoS marking/priority settings of the end points as per the defined policy and should support Rate Limiting - Policing and/or shaping	Please amend the clause: All ports on Switch must provide encryption of traffic ie. MAC Sec Encryption (802.1AE) in hardware or should support Port security and should support QoS marking/priority settings of the end points as per the defined policy and should support Rate Limiting - Policing and/or shaping. Justification: MACsec is a point to point layer 1 encryption, in order to run Macsec across the campus, it's a prerequisite that MACsec is supported across all platforms (access to core). Additionally Macsec devices are more expensive as they require a hardware based encryption engine inbuilt to carry out the line rate layer 1 encryption of traffic. Such requirements are an overkill within a campus environment, even in high security centric deployments, MACsec is only run over the links where the fibre is traversing/ deployed in area which is outside the administration/ control of the enterprise. Also	No Change
22	Page no 23	4. Technical Specification and Compliance Network Switch Point#8	All ports on Switch must provide encryption of traffic ie. MAC Sec Encryption (802.1AE) in hardware and should support QoS marking/priority settings of the end points as per the defined policy and should support Rate Limiting - Policing and/or shaping	Please amend the clause: All ports on Switch must provide encryption of traffic ie. MAC Sec Encryption (802.1AE) in hardware or should support Port security and should support QoS marking/priority settings of the end points as per the defined policy and should support Rate Limiting - Policing and/or shaping Justification: MACsec is a point to point layer 1 encryption, in order to run Macsec across the campus, it's a prerequisite that MACsec is supported across all platforms (access to core). Additionally Macsec devices are more expensive as they require a hardware based encryption engine inbuilt to carry out the line rate layer 1 encryption of traffic. Such requirements are an overkill within a campus environment, even in high security centric deployments, MACsec is only run over the links where the fibre is traversing/ deployed in area which is outside the administration/ control of the enterprise.	Same as query#21

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
23	24	5. Project Timelines & Payment Milestones	Supply, Installation and Commissioning : 1) Supply of complete Infrastructure as per Financial Proposal : T+ 6 : 20 % of the Work Order Value 2) Installation, commissioning all Infrastructure and system UAT/Verification : T+ 8 : 20 % of the Work Order Value Final Acceptance Test and Go-live 3) Integration with existing infra/solution and FAT of all the Installed, Commissioned Infrastructure and Go-Live (System is in usable state). : T+10 : 50 % of the Work Order Value Support and Maintenance 4) Support and Maintenance for 60 months (from the date of golive) : Go-Live + 60 Months : 10 % of the Work Order Value (2% per year from the date of Go-Live) which shall be paid quarterly i.e. 0.5% per quarter	Requesting to amend this clause as follows: Supply, Installation and Commissioning : 1) Supply of complete Infrastructure as per Financial Proposal : T+ 10 : 60% of the Work Order Value 2) Installation, commissioning all Infrastructure and system UAT/Verification : T+ 16 : 30 % of the Work Order Value Final Acceptance Test and Go-live 3) Integration with existing infra/solution and FAT of all the Installed, Commissioned Infrastructure and Go-Live (System is in usable state). : T+20 : 5 % of the Work Order Value Support and Maintenance 4) Support and Maintenance for 60 months (from the date of golive) : Go-Live + 60 Months: 5 % of the Work Order Value (1% per year from the date of Go-Live) which shall be paid quarterly i.e. 0.25% per quarter Justification: due to Global Memory /HDD/SSD/NVME Drives shortage overall OEM delivery are Impacted . We are committed to maintaining high-quality standards. Extending the delivery schedule will allow sufficient time for thorough testing, inspection, and quality assurance, ensuring long-term performance and reliability.	Refer Corrigendum-2 Sn#1, 2 & 3
24	24	6. Service Level Agreement (SLA) & Penalties	6.1 SLAs for Supply, Installation and Commissioning of Infrastructure:	Requesting to amend this clause as follows: Additional: Overall capping for the penalty will be 5%.	No Change

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
25	25	6. Service Level Agreement (SLA) & Penalties	6.2 SLA for support Service Type of Incident: Target: Penalty: Measurement Tool/Method Critical P1 Incidents (Critical System Down): Response: <=15 mins, Resolution: <= 4 hrs : No Penalty Response: >15 mins to <= 30 mins, Resolution: > 4 hrs to <= 6 hrs: 0.5% of the quarterly payment for every unattended/ unresolved call Response: >30 mins to <= 60 min, Resolution: >6 hrs to <= 8 hrs: 1% of the quarterly payment for every unattended/ unresolved call Response: > 60 min, Resolution: > 8 hrs: 2% of the quarterly payment for every unattended/ unresolved call for delay of every 15 minutes in Response and 2 hours in Resolution	Requesting to amend this clause as follows: Type of Incident: Target: Penalty: Measurement Tool/Method Critical P1 Incidents (Critical System Down): Response: <=2 Hrs., Resolution: <= 6 hrs : No Penalty Response: >2 Hrs to <= 3 Hrs, Resolution: > 6 hrs to <= 8 hrs: 0.5% of the quarterly payment for every unattended/ unresolved call Response: > 3 Hrs to <= 4 Hrs, Resolution: >8 hrs to <= 12 hrs: 1% of the quarterly payment for every unattended/ unresolved call Response: > 4 Hrs, Resolution: > 12 hrs: 2% of the quarterly payment for every unattended/ unresolved call for delay of every 15 minutes in Response and 2 hours in Resolution	No Change
26	26	6. Service Level Agreement (SLA) & Penalties	6.2 SLA for support Service Type of Incident: Target: Penalty: Measurement Tool/Method High P2 Incidents (Major Functionality Impaired) Response: <=30 mins, Resolution: <= 8 hrs: No Penalty Response: >30 mins to <= 60 mins, Resolution: > 8 hrs to <= 12 hrs: 0.5% of the quarterly payment for every unattended/ unresolved call Response: >60 mins to <= 90 mins, Resolution: 12 hrs to <= 16 hrs: 1% of the quarterly payment for every unattended/ unresolved call Response: > 90 min, Resolution: > 16 hrs: 2% of the quarterly payment for every unattended/ unresolved call for delay of every 30 minutes in Response and 4 hours in Resolution	Requesting to amend this clause as follows: Type of Incident: Target: Penalty: Measurement Tool/Method High P2 Incidents (Major Functionality Impaired) Response: <=4 hrs, Resolution: <= 12 hrs: No Penalty Response: >4 hrs to <= 6 hrs, Resolution: > 12 hrs to <= 24 hrs: 0.5% of the quarterly payment for every unattended/ unresolved call Response: >6 hrs to <= 8 hrs, Resolution: 24 hrs to <= 36 hrs: 1% of the quarterly payment for every unattended/ unresolved call Response: > 8 hrs, Resolution: > 36 hrs: 2% of the quarterly payment for every unattended/ unresolved call for delay of every 30 minutes in Response and 4 hours in Resolution	No Change

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
27	26	6. Service Level Agreement (SLA) & Penalties	6.2 SLA for support Service Type of Incident: Target: Penalty: Measurement Tool/Method Medium P3 Incidents (Minor Impact) Response: <= 2 hrs, Resolution: <= 24 hrs: No Penalty Response: >2 hrs to <= 4 hrs, Resolution: 24 hrs to <= 36 hrs: 0.5% of the quarterly payment for every unattended/ unresolved call Response: >4 hrs to <= 6 hrs, Resolution: 36 hrs to <= 48 hrs: 1% of the quarterly payment for every unattended/ unresolved call Response: >6 hrs, Resolution: > 48 hrs: 2% of the quarterly payment for every unattended/ unresolved call for delay of every 2 hrs in Response and 12 hours in Resolution	Requesting to amend this clause as follows: Type of Incident: Target: Penalty: Measurement Tool/Method Medium P3 Incidents (Minor Impact) Response: <= 6 hrs, Resolution: <= 48 hrs: No Penalty Response: >6 hrs to <= 8 hrs, Resolution: 48 hrs to <= 72 hrs: 0.5% of the quarterly payment for every unattended/ unresolved call Response: >8 hrs to <= 12 hrs, Resolution: 72 hrs to <= 84 hrs: 1% of the quarterly payment for every unattended/ unresolved call Response: >12 hrs, Resolution: > 84 hrs: 2% of the quarterly payment for every unattended/ unresolved call for delay of every 2 hrs in Response and 12 hours in Resolution	No Change
28	27	6. Service Level Agreement (SLA) & Penalties	Note: 2. Penalties shall be capped at a maximum of 50% of the Value of Quarterly invoice. In case of default, MPSEDC may choose to terminate the Contract.	Requesting to amend this clause as follows: Note: 2. Penalties shall be capped at a maximum of 10% of the Value of Quarterly invoice. In case of default, MPSEDC may choose to terminate the Contract.	No Change
29	27	6. Service Level Agreement (SLA) & Penalties	Additional	Requesting to please provide the criticality level of the component as per the Bill of Quantity. Based on this, the SLA will be applicable.	No Change
30	12	Eligibility / Qualification Criteria	Additional	Proposed Clause to be Added in PQ Both the OEM and the Bidder should not be blacklisted or debarred by any Central Government, State Government, or any Government PSU in India as on the date of bid submission. Justification: The clause ensures participation of credible and compliant OEMs and bidders, reduces legal and reputational risk, and promotes transparency and fair competition in the procurement process.	No Change
31	16	Scope of work	Resiliency: Data resiliency with FTT- 1/RF-2, with future support for FTT-2/RF-3.	Modification : Resiliency: Data resiliency with FTT- 2/RF-3 to support SLA & availability of Critical SDC workload , Solution must be offered with 1Ms response time with asked IOPS of 100000 70:30 R:W- 16KB Block size Justification: Resiliency for Critical SDC workload, FTT2/RF3 will provide 2 Node failover resiliency , This will ensure availability required for SDC critical workload	Same as query#8

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32	20	Technical Specifications & Compliances	Offered Solution shall be a next generation software defined platform which shall offer the functionality of Hyper-converged with independent scaling of both Storage and Compute without any downtime. The solution should provide enterprise-class storage services using latest x86 server infrastructures without dependence on a separate Storage Area Network Storage & associated components such as SAN Switches & HBAs. Offered HCI system should be a software-centric system for delivering compute, storage and networking resources in a tightly integrated system	Modification Request: Offered Solution shall be a next generation software defined platform which shall offer the functionality of Hyper-converged with independent scaling of both Storage and Compute without any downtime. The solution should provide enterprise-class storage services using latest x86 server infrastructures without dependence on a separate Storage Area Network Storage & associated components such as SAN Switches & HBAs. Offered HCI system should be a software-centric system for delivering compute, storage and networking resources in a tightly integrated HCI solution to deliver asked HCI services Justification: In HCI solution offering, different OEM have different architecture to deliver Virtual machine with Performance Integrated with compute, storage and networking resources.	Same as query#9
33	21	Technical Specifications & Compliances	Resource requirement: HCI Solution should be provided with N+1 configuration having maximum 12 nodes Hybrid cluster, each node providing latest Intel processors/ AMD processors with base frequency of 2.1GHz or higher and DDR5 Memory. Solution must be proposed with at least 1500 or higher cores with minimum 1300 usable cores, 6 TB or higher memory and with minimum 500 TB usable storage without considering any deduplication, compression, erasure coding or any saving techniques benefits. Complete solution must be proposed with RF2/FTT1 i.e. two copies of data across the cluster. Each server/node should be configured with minimum of one Quad Port 10/25G network ports along having required 25G SFPs/DAC Cables for uplink connectivity	Resource requirement: HCI Solution should be provided with N+1 configuration having maximum 12 nodes Hybrid cluster, each node providing latest Intel processors/ AMD processors with base frequency of 2.1GHz or higher and DDR5 Memory. Solution must be proposed with at least 1500 or higher cores with minimum 1300 usable cores, 6 TB or higher memory and with minimum 500 TB usable storage without considering any deduplication, compression, erasure coding or any saving techniques benefits. Complete solution must be proposed with RF3/FTT2 i.e. two copies of data across the cluster. Each server/node should be configured with minimum of one Quad Port 10/25G network ports along having required 25G SFPs/DAC Cables for uplink connectivity Justification: Resiliency for Critical SDC workload & to ensure SLA with minimum 2 Node Failure architecture	Same as query#10

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34	21	Technical Specifications & Compliances	The proposed HCI solution should be a factory shipped engineered & integrated appliance. The HCI solution should support SSD/NLSAS Drives. The HCI solution should support scalability up to 32 nodes in a single cluster. Each server node should have dedicated redundant hot swap power supply, cooling fans. There should not be any separate disaggregated components in the proposed HCI solution i.e. The 3 Major components of the HCI node i.e. Compute, RAM and Storage should be physically inside the node, no resource should be provisioned from outside the HCI Node.	Modification: The proposed HCI solution should be a factory shipped engineered & integrated appliance. The HCI solution should be offered with NVMe Drives only. The HCI solution should support scalability up to 16 nodes in a single cluster. Each server node should have dedicated redundant hot swap power supply, cooling fans. There should not be any separate disaggregated components in the proposed HCI solution i.e. The 3 Major components of the HCI node i.e. Compute, RAM and Storage should be physically inside the node, no resource should be provisioned from outside the HCI Node or as per OEM architecture ensuring full software defined architecture delivering required performance & IOPS. Justification: 1. Please Change asked Drives to NVMe Drives only , as NL SAS/ SSD /SAS have no Future road Map , Considering this is SDC Critical Environmmnet any componnet which is going to be obsolute in near future should not be considered . 2. Change asked architecture with NVMe Drives only , This will ensure performance, abalability & high resileancy with Latest NVMe architectire. 2. Request Change in asked Cluster Scalability as 16/32 Nodes , since 32 Node in single cluster as SDS /HCI solution is not recommended from Management & for future Upgrade & Updated point of view.	Same as query#11
35	21	Technical Specifications & Compliances	The HCI solution shall support scaling Compute and Storage capacity and performance linearly by addition of nodes. VMs on existing nodes shall get the storage performance & capacity that gets enhanced by the addition of new HCI nodes in which HCI solution must provide on the fly change of ESE(Enterprise Storage Efficiency)- Deduplication/Compression for workload without any visible impact on storage and their operations and the bidder must ensure minimum 100,000 IOPS from the proposed HCI solution, assuming 8kb block size, and 70:30 read: write ratio.	Modification: The HCI solution shall support scaling Compute and Storage capacity and performance linearly by addition of nodes. VMs on existing nodes shall get the storage performance & capacity that gets enhanced by the addition of new HCI nodes in which HCI solution must provide on the fly change of ESE(Enterprise Storage Efficiency)- Deduplication/Compression for workload without any visible impact on storage and their operations and the bidder must ensure minimum 100,000 IOPS from the proposed HCI solution, assuming 8kb block size, and 70:30 read: write ratio ensuring less than 1ms response time . Justification: Since SDC application & DB Requirement is Critical from Performance POV. Pls define Response time to ensure Performance delivery & architecture is offered	Same as query#12

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
36	21	Technical Specifications & Compliances	The HCI software should pool all HDDs from all the nodes in the cluster to present a single storage resource pool to all server nodes in the cluster. The HCI storage should be a scale-out distributed storage. Proposed HCI solution must have a primary copy of the VM data locally to the Server/Node to offer faster read and write and secondary copy must be distributed on the other HCI Nodes. The intelligent software should consolidate the resources from each server node into a shared resource pool, delivering a high degree of flexibility while also simplifying management.	Modification: The HCI software should pool all NVMe HDDs from all the nodes in the cluster to present a single storage resource pool to all server nodes in the cluster. The HCI storage should be a scale-out distributed storage. Proposed HCI solution must have a primary copy of the VM data locally to the Server/Node to offer faster read and write and secondary copy must be distributed on the other HCI Nodes or as per OEM architecture ensuring less than 1ms sustainable response time. The intelligent software should consolidate the resources from each server node into a shared resource pool, delivering a high degree of flexibility while also simplifying management. Justification: Please change asked Disk type of Storage System to : NVMe. Kindly modify this clause for participation and better protection of investment.	Same as query#13
37	21	Technical Specifications & Compliances	The solution should automatically rebalance data to maintain balanced utilization of storage across the HCI nodes. When storage capacity is scaled up or scaled out, the HCI nodes must automatically redistribute data equally across all nodes equally without migrating VMs. Hypervisor software must provide Data at rest encryption with Native KMS which protects unauthorized data access.	Modification: The solution should automatically rebalance data to maintain balanced utilization of storage across the HCI nodes. When storage capacity is scaled up or scaled out, the HCI nodes must automatically redistribute data equally across all nodes equally without migrating VMs. Hypervisor / HCI solution must provide Data at rest encryption with Native KMS which protects unauthorized data access. Justification: Please change asked Hypervisor Based encryption to HCI Solution based Data at rest encryption - Option for Hardware based Encryption, as per OEM architecture to deliver Data at rest encryption feature	Same as query#14
38	21	Technical Specifications & Compliances	HCI solution should support Block, File and Object (S3) natively or using third party solution from Day1. Solution should support file storage supporting NFS v3/v4 and SMB 2.0/3.0 for Linux and Windows Guest with unlimited shares v with Active directory/LDAP	This is Restrictive Clauses, Please Remove this Clause / Modify : HCI solution should support Block, File natively or using third party solution from Day1. Solution should support file storage supporting NFS v3/v4/SMB 2.0/3.0 for Linux and Windows Guest with unlimited shares v with Active directory/LDAP Justification: This is Restrictive Clauses, This point is OEM specific to one OEM Request you to remove the ask of S3 protocol for wider participation of OEM	Same as query#15
39	21	Technical Specifications & Compliances	The solution quoted shall provide VM level/ Native Storage replication from a Primary site for planned DR Site strategy and vice-versa. If required, bidder can leverage existing replication tool.	The solution quoted shall provide VM level/ Native Storage replication from a Primary site for planned and running DC & DR Site strategy and vice-versa. The proposed solution should have a capability to integrate with existing deployed cloud platform running Visualized and physical hardware. If required, bidder can leverage existing replication tool.	Same as query#16

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
40	21	Technical Specifications & Compliances	Solution should support Virtual Network visibility with application-centric protection from network threats and automation of common networking operations	Modification: Solution should support Virtual Network visibility with application-centric protection from network threats with automation of common existing network operations, security appliances & firewall, load balancer, backup solution. Justification: for seamless integration with existing network, load balancer and security devices. please include the ask modification..	Same as query#17
41	22	Technical Specifications & Compliances	The management tool should be able to provide global resource pooling and policy management to enable policy-based automation. The proposed management solution should provide proactive security & software advisory alerts and should outline the fixes required to address the issues and the management software should provide a mobile app for monitoring & viewing the resources being managed in the infrastructure	Modification: The management tool should be able to provide global resource pooling and policy management to enable policy-based automation. The proposed management solution should provide proactive security & software advisory alerts and should outline the fixes required to address the issues . Justification: "Security Breach" Please Remove the ask of Mobile app for monitoring & viewing the resources being managed in the infrastructure, as allowing Mobile app for monitoring & viewing the resources will be point of security breach in environment & will have security impact in SDC environment "	Same as query#18
42	22	Technical Specifications & Compliances	Switch should support minimum 1000 VRF instances with route leaking functionality, minimum 400K IPv4 LPM routes, min 16K Multicast routes, 3. minimum 90K no. of MAC addresses, minimum 32 MB Packet Buffer with intelligent shared-memory buffered architecture to simplify the system buffer management and queuing implementation.	Please amend the clause as: Switch should support minimum 256 VRF instances with route leaking functionality, minimum 120K IPv4 LPM routes, min 7K Multicast routes, 3. minimum 90K no. of MAC addresses, minimum 32 MB Packet Buffer with intelligent shared-memory buffered architecture to simplify the system buffer management and queuing implementation. Justification: 256 VRF are more than sufficient and higher VRF required in ISP environment which supports multiple tenant.7K multicast routes more than sufficient for current deployment scenario, not more than 100 Multicast will be used.	Same as query#19
43	23	Technical Specifications & Compliances	Switch should support VXLAN to achieve Network Virtualisation, along with VXLAN and EVPN symmetric IRB. The switch should support BGP EVPN Route Type 2, Type 4 and Route Type 5 for the overlay control plane	Please amend the clause: Switch should support VXLAN to achieve Network Virtualisation, along with VXLAN and EVPN symmetric IRB. The switch should support BGP EVPN Route Type 2, Type 3/Type4 and Route Type 5 for the overlay control plane Justification: request to amend the clause for wider participation.	Same as query#20

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
44	23		All ports on Switch must provide encryption of traffic i.e. MAC Sec Encryption (802.1AE) in hardware and should support QoS marking/priority settings of the end points as per the defined policy and should support Rate Limiting - Policing and/or shaping	Please amend the clause: All ports on Switch must provide encryption of traffic i.e. MAC Sec Encryption (802.1AE) in hardware or should support Port security and should support QoS marking/priority settings of the end points as per the defined policy and should support Rate Limiting - Policing and/or shaping. Justification: MACsec is a point to point layer 1 encryption, in order to run Macsec across the campus, it's a prerequisite that MACsec is supported across all platforms (access to core). Additionally Macsec devices are more expensive as they require a hardware based encryption engine inbuilt to carry out the line rate layer 1 encryption of traffic. Such requirements are an overkill within a campus environment, even in high security centric deployments, MACsec is only run over the links where the fibre is traversing/deployed in area which is outside the administration/control of the enterprise. Also	Same as query#21
45	23	Technical Specifications & Compliances	All ports on Switch must provide encryption of traffic i.e. MAC Sec Encryption (802.1AE) in hardware and should support QoS marking/priority settings of the end points as per the defined policy and should support Rate Limiting - Policing and/or shaping	Please amend the clause: All ports on Switch must provide encryption of traffic i.e. MAC Sec Encryption (802.1AE) in hardware or should support Port security and should support QoS marking/priority settings of the end points as per the defined policy and should support Rate Limiting - Policing and/or shaping Justification: MACsec is a point to point layer 1 encryption, in order to run Macsec across the campus, it's a prerequisite that MACsec is supported across all platforms (access to core). Additionally Macsec devices are more expensive as they require a hardware based encryption engine inbuilt to carry out the line rate layer 1 encryption of traffic. Such requirements are an overkill within a campus environment, even in high security centric deployments, MACsec is only run over the links where the fibre is traversing/deployed in area which is outside the administration/control of the enterprise.	Same as query#21
46	Page no 16	Scope of work	Resiliency: Data resiliency with FTT- 1/RF-2, with future support for FTT-2/RF-3.	Modification : Resiliency: Data resiliency with FTT- 2/RF-3 to support SLA & availability of Critical SDC workload , Solution must be offered with 1Ms response time with asked IOPS of 100000 70:30 R:W- 16KB Block size Justification: Resiliency for Critical SDC workload, FTT2/RF3 will provide 2 Node failover resiliency , This will ensure availability required for SDC critical workload	Same as query#8

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
47	Page no 20	Technical Specifications & Compliances	Offered Solution shall be a next generation software defined platform which shall offer the functionality of Hyper-converged with independent scaling of both Storage and Compute without any downtime. The solution should provide enterprise-class storage services using latest x86 server infrastructures without dependence on a separate Storage Area Network Storage & associated components such as SAN Switches & HBAs. Offered HCI system should be a software-centric system for delivering compute, storage and networking resources in a tightly integrated system	Modification Request: Offered Solution shall be a next generation software defined platform which shall offer the functionality of Hyper-converged with independent scaling of both Storage and Compute without any downtime. The solution should provide enterprise-class storage services using latest x86 server infrastructures without dependence on a separate Storage Area Network Storage & associated components such as SAN Switches & HBAs. Offered HCI system should be a software-centric system for delivering compute, storage and networking resources in a tightly integrated HCI solution to deliver asked HCI services Justification: In HCI solution offering , different OEM have different architecture to deliver Virtual machine with Performance Integrated with compute, storage and networking resources .	Same as query #9
48	Page no 21		Resource requirement: HCI Solution should be provided with N+1 configuration having maximum 12 nodes Hybrid cluster, each node providing latest Intel processors/ AMD processors with base frequency of 2.1GHz or higher and DDR5 Memory. Solution must be proposed with at least 1500 or higher cores with minimum 1300 usable cores, 6 TB or higher memory and with minimum 500 TB usable storage without considering any deduplication, compression, erasure coding or any saving techniques benefits. Complete solution must be proposed with RF2/FTT1 i.e. two copies of data across the cluster. Each server/node should be configured with minimum of one Quad Port 10/25G network ports along having required 25G SFPs/DAC Cables for uplink connectivity	Resource requirement: HCI Solution should be provided with N+1 configuration having maximum 12 nodes Hybrid cluster, each node providing latest Intel processors/ AMD processors with base frequency of 2.1GHz or higher and DDR5 Memory. Solution must be proposed with at least 1500 or higher cores with minimum 1300 usable cores, 6 TB or higher memory and with minimum 500 TB usable storage without considering any deduplication, compression, erasure coding or any saving techniques benefits. Complete solution must be proposed with RF3/FTT2 i.e. two copies of data across the cluster. Each server/node should be configured with minimum of one Quad Port 10/25G network ports along having required 25G SFPs/DAC Cables for uplink connectivity Justification: Resiliency for Critical SDC workload & to ensure SLA with minimum 2 Node Failover architecture	Same as query #10

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
49	Page no 21	Technical Specifications & Compliances	The proposed HCI solution should be a factory shipped engineered & integrated appliance. The HCI solution should support SSD/NLSAS Drives. The HCI solution should support scalability up to 32 nodes in a single cluster. Each server node should have dedicated redundant hot swap power supply, cooling fans. There should not be any separate disaggregated components in the proposed HCI solution i.e. The 3 Major components of the HCI node i.e. Compute, RAM and Storage should be physically inside the node, no resource should be provisioned from outside the HCI Node.	Modification: The proposed HCI solution should be a factory shipped engineered & integrated appliance. The HCI solution should be offered with NVMe Drives only. The HCI solution should support scalability up to 16 nodes in a single cluster. Each server node should have dedicated redundant hot swap power supply, cooling fans. There should not be any separate disaggregated components in the proposed HCI solution i.e. The 3 Major components of the HCI node i.e. Compute, RAM and Storage should be physically inside the node, no resource should be provisioned from outside the HCI Node or as per OEM architecture ensuring full software defined architecture delivering required performance & IOPS. Justification: 1. Please Change asked Drives to NVMe Drives only , as NL SAS/ SSD /SAS have no Future road Map , Considering this is SDC Critical Environmnet any componnet which is going to be obsolute in near future should not be considered . 2. Change asked architecture with NVMe Drives only , This will ensure performance, abalability & high resileancy with Latest NVMe architctire. 2. Request Change in asked Cluster Scalability as 16/32 Nodes , since 32 Node in single cluster as SDS /HCI solution is not recommended from Management & for future Upgrade & Updated point of view.	Same as query#11
50	Page no 21	Technical Specifications & Compliances	The HCI solution shall support scaling Compute and Storage capacity and performance linearly by addition of nodes. VMs on existing nodes shall get the storage performance & capacity that gets enhanced by the addition of new HCI nodes in which HCI solution must provide on the fly change of ESE(Enterprise Storage Efficiency)- Deduplication/Compression for workload without any visible impact on storage and their operations and the bidder must ensure minimum 100,000 IOPS from the proposed HCI solution, assuming 8kb block size, and 70:30 read: write ratio.	Modification: The HCI solution shall support scaling Compute and Storage capacity and performance linearly by addition of nodes. VMs on existing nodes shall get the storage performance & capacity that gets enhanced by the addition of new HCI nodes in which HCI solution must provide on the fly change of ESE(Enterprise Storage Efficiency)- Deduplication/Compression for workload without any visible impact on storage and their operations and the bidder must ensure minimum 100,000 IOPS from the proposed HCI solution, assuming 8kb block size, and 70:30 read: write ratio ensuring less than 1ms response time . Justification: Since SDC application & DB Requirement is Critical from Performance POV. Pls define Response time to ensure Performance delivery & architecture is offered	Same as query#12

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
51	Page no 21	Technical Specifications & Compliances	The HCI software should pool all HDDs from all the nodes in the cluster to present a single storage resource pool to all server nodes in the cluster. The HCI storage should be a scale-out distributed storage. Proposed HCI solution must have a primary copy of the VM data locally to the Server/Node to offer faster read and write and secondary copy must be distributed on the other HCI Nodes. The intelligent software should consolidate the resources from each server node into a shared resource pool, delivering a high degree of flexibility while also simplifying management.	Modification: The HCI software should pool all NVMe HDDs from all the nodes in the cluster to present a single storage resource pool to all server nodes in the cluster. The HCI storage should be a scale-out distributed storage. Proposed HCI solution must have a primary copy of the VM data locally to the Server/Node to offer faster read and write and secondary copy must be distributed on the other HCI Nodes or as per OEM architecture ensuring less than 1ms sustainable response time. The intelligent software should consolidate the resources from each server node into a shared resource pool, delivering a high degree of flexibility while also simplifying management. Justification: Please change asked Disk type of Storage System to : NVMe. Kindly modify this clause for participation and better protection of investment.	Same as query#13
52	Page no 21	Technical Specifications & Compliances	The solution should automatically rebalance data to maintain balanced utilization of storage across the HCI nodes. When storage capacity is scaled up or scaled out, the HCI nodes must automatically redistribute data equally across all nodes equally without migrating VMs. Hypervisor software must provide Data at rest encryption with Native KMS which protects unauthorized data access.	Modification: The solution should automatically rebalance data to maintain balanced utilization of storage across the HCI nodes. When storage capacity is scaled up or scaled out, the HCI nodes must automatically redistribute data equally across all nodes equally without migrating VMs. Hypervisor / HCI solution must provide Data at rest encryption with Native KMS which protects unauthorized data access. Justification: Please change asked Hypervisor Based encryption to HCI Solution based Data at rest encryption - Option for Hardware based Encryption, as per OEM architecture to deliver Data at rest encryption feature	Same as query#14
53	Page no 21	Technical Specifications & Compliances	HCI solution should support Block, File and Object (S3) natively or using third party solution from Day1. Solution should support file storage supporting NFS v3/v4 and SMB 2.0/3.0 for Linux and Windows Guest with unlimited shares v with Active directory/LDAP	This is Restrictive Clauses, Please Remove this Clause / Modify : HCI solution should support Block, File natively or using third party solution from Day1. Solution should support file storage supporting NFS v3/v4/SMB 2.0/3.0 for Linux and Windows Guest with unlimited shares v with Active directory/LDAP Justification: This is Restrictive Clauses, This point is OEM specific to one OEM Request you to remove the ask of S3 protocol for wider participation of OEM	Same as query#15
54	Page no 21	Technical Specifications & Compliances	The solution quoted shall provide VM level/ Native Storage replication from a Primary site for planned DR Site strategy and vice-versa. If required, bidder can leverage existing replication tool.	The solution quoted shall provide VM level/ Native Storage replication from a Primary site for planned and running DC & DR Site strategy and vice-versa. The proposed solution should have a capability to integrate with existing deployed cloud platform running Visualized and physical hardware. If required, bidder can leverage existing replication tool.	Same as query#16

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
55	Page no 21	Technical Specifications & Compliances	Solution should support Virtual Network visibility with application-centric protection from network threats and automation of common networking operations	Modification: Solution should support Virtual Network visibility with application-centric protection from network threats with automation of common existing network operations, security appliances & firewall, load balancer, backup solution. Justification: for seamless integration with existing network, load balancer and security devices. please include the ask modification..	Same as query#17
56	Page no 22	Technical Specifications & Compliances	The management tool should be able to provide global resource pooling and policy management to enable policy-based automation. The proposed management solution should provide proactive security & software advisory alerts and should outline the fixes required to address the issues and the management software should provide a mobile app for monitoring & viewing the resources being managed in the infrastructure	Modification: The management tool should be able to provide global resource pooling and policy management to enable policy-based automation. The proposed management solution should provide proactive security & software advisory alerts and should outline the fixes required to address the issues . Justification: "Security Breach" Please Remove the ask of Mobile app for monitoring & viewing the resources being managed in the infrastructure, as allowing Mobile app for monitoring & viewing the resources will be point of security breach in environment & will have security impact in SDC environment "	Same as query#18
57	Page no 22	Technical Specifications & Compliances	Switch should support minimum 1000 VRF instances with route leaking functionality, minimum 400K IPv4 LPM routes, min 16K Multicast routes, 3. minimum 90K no. of MAC addresses, minimum 32 MB Packet Buffer with intelligent shared-memory buffered architecture to simplify the system buffer management and queuing implementation.	Please amend the clause as: Switch should support minimum 256 VRF instances with route leaking functionality, minimum 120K IPv4 LPM routes, min 7K Multicast routes, 3. minimum 90K no. of MAC addresses, minimum 32 MB Packet Buffer with intelligent shared-memory buffered architecture to simplify the system buffer management and queuing implementation. Justification: 256 VRF are more than sufficient and higher VRF required in ISP environment which supports multiple tenant. 7K multicast routes more than sufficient for current deployment scenario, not more than 100 Multicast will be used.	Same as query#19
58	Page no 23	Technical Specifications & Compliances	Switch should support VXLAN to achieve Network Virtualisation, along with VXLAN and EVPN symmetric IRB. The switch should support BGP EVPN Route Type 2, Type 4 and Route Type 5 for the overlay control plane	Please amend the clause: Switch should support VXLAN to achieve Network Virtualisation, along with VXLAN and EVPN symmetric IRB. The switch should support BGP EVPN Route Type 2, Type 3/Type4 and Route Type 5 for the overlay control plane Justification: request to amend the clause for wider participation.	Same as query#20

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
59	Page no 23		All ports on Switch must provide encryption of traffic ie. MAC Sec Encryption (802.1AE) in hardware and should support QoS marking/priority settings of the end points as per the defined policy and should support Rate Limiting - Policing and/or shaping	Please amend the clause: All ports on Switch must provide encryption of traffic ie. MAC Sec Encryption (802.1AE) in hardware or should support Port security and should support QoS marking/priority settings of the end points as per the defined policy and should support Rate Limiting - Policing and/or shaping. Justification: MACsec is a point to point layer 1 encryption, in order to run Macsec across the campus, it's a prerequisite that MACsec is supported across all platforms (access to core). Additionally Macsec devices are more expensive as they require a hardware based encryption engine inbuild to carry out the line rate layer 1 encryption of traffic. Such requirements are an overkill within a campus environment, even in high security centric deployments, MACsec is only run over the links where the fibre is traversing/deployed in area which is outside the administration/control of the enterprise. Also	Same as query#21
60	Page no 23	Technical Specifications & Compliances	All ports on Switch must provide encryption of traffic ie. MAC Sec Encryption (802.1AE) in hardware and should support QoS marking/priority settings of the end points as per the defined policy and should support Rate Limiting - Policing and/or shaping	Please amend the clause: All ports on Switch must provide encryption of traffic ie. MAC Sec Encryption (802.1AE) in hardware or should support Port security and should support QoS marking/priority settings of the end points as per the defined policy and should support Rate Limiting - Policing and/or shaping Justification: MACsec is a point to point layer 1 encryption, in order to run Macsec across the campus, it's a prerequisite that MACsec is supported across all platforms (access to core). Additionally Macsec devices are more expensive as they require a hardware based encryption engine inbuild to carry out the line rate layer 1 encryption of traffic. Such requirements are an overkill within a campus environment, even in high security centric deployments, MACsec is only run over the links where the fibre is traversing/deployed in area which is outside the administration/control of the enterprise.	Same as query#21
61	Page no 13	Experience in Data Centre or Disaster Recovery site projects of IT infra implementation and O&M	The bidder must have successfully completed an assignment of similar nature involving the Supply, Installation, Configuring, and Maintenance of Data Centre or	We request to consider the experience credentials from Public Listed Companies eligibility criteria under the RFP. Additionally, we request confirmation on the inclusion of experience pertaining to Cloud Services and Cloud Disaster Recovery (DR) Services for eligibility evaluation. Inclusion of the above would help broaden participation and is likely to ensure healthy competition in the RFP process.	No Change

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
62	Page no 13	Experience in Data Centre or Disaster Recovery site projects of IT infra implementation and O&M	he bidder must have successfully completed an assignment of similar nature involving the Supply, Installation, Configuring, and Maintenance of Data Centre or Disaster Recovery site projects of IT infra implementation and O&M within India. Provided that project shall not be of the period prior to 1st April 2020 and must have undertaken for any of the following: A) State or Central Government OR BJ State or Central PSU OR C) Scheduled Commercial Banks (as per RBI's latest notification) The below mentioned order values are applicable for the above criteria: The bidder should have undertaken One project of a similar nature with a minimum value of INR 14 Crores. OR Two projects of a similar nature each with a minimum value of INR 10 Crores. OR Three projects of a similar nature, each with a minimum value of INR 07 Crores	We request to consider the experience credentials from Public Listed Companies eligibility criteria under the RFP. Additionally, we request confirmation on the inclusion of experience pertaining to Cloud Services and Cloud Disaster Recovery (DR) Services for eligibility evaluation. Inclusion of the above would help broaden participation and is likely to ensure healthy competition in the RFP process.	No Change
63	Page no 16	Scope of work	Resiliency: Data resiliency with FTT-1/RF-2, with future support for FTT-2/RF-3.	Modification : Resiliency: Data resiliency with FTT-2/RF-3 to support SLA & availability of Critical SDC workload , Solution must be offered with 1Ms response time with asked IOPS of 100000 70:30 R:W- 16KB Block size Justification: Resiliency for Critical SDC workload, FTT2/RF3 will provide 2 Node failover resiliency , This will ensure availability required for SDC critical workload	Same as query #8

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
64	Page no 20	Technical Specifications & Compliances	Offered Solution shall be a next generation software defined platform which shall offer the functionality of Hyper-converged with independent scaling of both Storage and Compute without any downtime. The solution should provide enterprise-class storage services using latest x86 server infrastructures without dependence on a separate Storage Area Network Storage & associated components such as SAN Switches & HBAs. Offered HCI system should be a software-centric system for delivering compute, storage and networking resources in a tightly integrated system	Modification Request: Offered Solution shall be a next generation software defined platform which shall offer the functionality of Hyper-converged with independent scaling of both Storage and Compute without any downtime. The solution should provide enterprise-class storage services using latest x86 server infrastructures without dependence on a separate Storage Area Network Storage & associated components such as SAN Switches & HBAs. Offered HCI system should be a software-centric system for delivering compute, storage and networking resources in a tightly integrated HCI solution to deliver asked HCI services Justification: In HCI solution offering, different OEM have different architecture to deliver Virtual machine with Performance Integrated with compute, storage and networking resources.	Same as query #9
65	Page no 21		Resource requirement: HCI Solution should be provided with N+1 configuration having maximum 12 nodes Hybrid cluster, each node providing latest Intel processors/ AMD processors with base frequency of 2.1GHz or higher and DDR5 Memory. Solution must be proposed with at least 1500 or higher cores with minimum 1300 usable cores, 6 TB or higher memory and with minimum 500 TB usable storage without considering any deduplication, compression, erasure coding or any saving techniques benefits. Complete solution must be proposed with RF2/FTT1 i.e. two copies of data across the cluster. Each server/node should be configured with minimum of one Quad Port 10/25G network ports along having required 25G SFPS/DAC Cables for uplink connectivity	Resource requirement: HCI Solution should be provided with N+1 configuration having maximum 12 nodes Hybrid cluster, each node providing latest Intel processors/ AMD processors with base frequency of 2.1GHz or higher and DDR5 Memory. Solution must be proposed with at least 1500 or higher cores with minimum 1300 usable cores, 6 TB or higher memory and with minimum 500 TB usable storage without considering any deduplication, compression, erasure coding or any saving techniques benefits. Complete solution must be proposed with RF3/FTT2 i.e. two copies of data across the cluster. Each server/node should be configured with minimum of one Quad Port 10/25G network ports along having required 25G SFPS/DAC Cables for uplink connectivity Justification: Resiliency for Critical SDC workload & to ensure SLA with minimum 2 Node Failover architecture	Same as query #10

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
66	Page no 21	Technical Specifications & Compliances	The proposed HCI solution should be a factory shipped engineered & integrated appliance. The HCI solution should support SSD/NLSAS Drives. The HCI solution should support scalability up to 32 nodes in a single cluster. Each server node should have dedicated redundant hot swap power supply, cooling fans. There should not be any separate disaggregated components in the proposed HCI solution i.e. The 3 Major components of the HCI node i.e. Compute, RAM and Storage should be physically inside the node, no resource should be provisioned from outside the HCI Node.	Modification: The proposed HCI solution should be a factory shipped engineered & integrated appliance. The HCI solution should be offered with NVMe Drives only. The HCI solution should support scalability up to 16 nodes in a single cluster. Each server node should have dedicated redundant hot swap power supply, cooling fans. There should not be any separate disaggregated components in the proposed HCI solution i.e. The 3 Major components of the HCI node i.e. Compute, RAM and Storage should be physically inside the node, no resource should be provisioned from outside the HCI Node or as per OEM architecture ensuring full software defined architecture delivering required performance & IOPS. Justification: 1. Please Change asked Drives to NVMe Drives only , as NL SAS/ SSD /SAS have no Future road Map , Considering this is SDC Critical Environmnet any componnet which is going to be obsolute in near future should not be considered . 2. Change asked architecture with NVMe Drives only , This will ensure performance, abalability & high resileancy with Latest NVMe architectire. 2. Request Change in asked Cluster Scalability as 16/32 Nodes , since 32 Node in single cluster as SDS /HCI solution is not recommended from Management & for future Upgrade & Updated point of view.	Same as query#11
67	Page no 21	Technical Specifications & Compliances	The HCI solution shall support scaling Compute and Storage capacity and performance linearly by addition of nodes. VMs on existing nodes shall get the storage performance & capacity that gets enhanced by the addition of new HCI nodes in which HCI solution must provide on the fly change of ESE(Enterprise Storage Efficiency)- Deduplication/Compression for workload without any visible impact on storage and their operations and the bidder must ensure minimum 100,000 IOPS from the proposed HCI solution, assuming 8kb block size, and 70:30 read: write ratio.	Modification: The HCI solution shall support scaling Compute and Storage capacity and performance linearly by addition of nodes. VMs on existing nodes shall get the storage performance & capacity that gets enhanced by the addition of new HCI nodes in which HCI solution must provide on the fly change of ESE(Enterprise Storage Efficiency)- Deduplication/Compression for workload without any visible impact on storage and their operations and the bidder must ensure minimum 100,000 IOPS from the proposed HCI solution, assuming 8kb block size, and 70:30 read: write ratio ensuring less than 1ms response time . Justification: Since SDC application & DB Requirement is Critical from Performance POV. Pls define Response time to ensure Performance delivery & architecture is offered	Same as query#12

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
68	Page no 21	Technical Specifications & Compliances	The HCI software should pool all HDDs from all the nodes in the cluster to present a single storage resource pool to all server nodes in the cluster. The HCI storage should be a scale-out distributed storage. Proposed HCI solution must have a primary copy of the VM data locally to the Server/Node to offer faster read and write and secondary copy must be distributed on the other HCI Nodes. The intelligent software should consolidate the resources from each server node into a shared resource pool, delivering a high degree of flexibility while also simplifying management.	Modification: The HCI software should pool all NVMe HDDs from all the nodes in the cluster to present a single storage resource pool to all server nodes in the cluster. The HCI storage should be a scale-out distributed storage. Proposed HCI solution must have a primary copy of the VM data locally to the Server/Node to offer faster read and write and secondary copy must be distributed on the other HCI Nodes or as per OEM architecture ensuring less than 1ms sustainable response time. The intelligent software should consolidate the resources from each server node into a shared resource pool, delivering a high degree of flexibility while also simplifying management. Justification: Please change asked Disk type of Storage System to : NVMe. Kindly modify this clause for participation and better protection of investment.	Same as query#13
69	Page no 21	Technical Specifications & Compliances	The solution should automatically rebalance data to maintain balanced utilization of storage across the HCI nodes. When storage capacity is scaled up or scaled out, the HCI nodes must automatically redistribute data equally across all nodes equally without migrating VMs. Hypervisor software must provide Data at rest encryption with Native KMS which protects unauthorized data access.	Modification: The solution should automatically rebalance data to maintain balanced utilization of storage across the HCI nodes. When storage capacity is scaled up or scaled out, the HCI nodes must automatically redistribute data equally across all nodes equally without migrating VMs. Hypervisor / HCI solution must provide Data at rest encryption with Native KMS which protects unauthorized data access. Justification: Please change asked Hypervisor Based encryption to HCI Solution based Data at rest encryption - Option for Hardware based Encryption, as per OEM architecture to deliver Data at rest encryption feature	Same as query#14
70	Page no 21	Technical Specifications & Compliances	HCI solution should support Block, File and Object (S3) natively or using third party solution from Day1. Solution should support file storage supporting NFS v3/v4 and SMB 2.0/3.0 for Linux and Windows Guest with unlimited shares v with Active directory/LDAP	This is Restrictive Clauses, Please Remove this Clause / Modify : HCI solution should support Block, File natively or using third party solution from Day1. Solution should support file storage supporting NFS v3/v4/SMB 2.0/3.0 for Linux and Windows Guest with unlimited shares v with Active directory/LDAP Justification: This is Restrictive Clauses, This point is OEM specific to one OEM Request you to remove the ask of S3 protocol for wider participation of OEM	Same as query#15
71	Page no 21	Technical Specifications & Compliances	The solution quoted shall provide VM level/ Native Storage replication from a Primary site for planned DR Site strategy and vice-versa. If required, bidder can leverage existing replication tool.	The solution quoted shall provide VM level/ Native Storage replication from a Primary site for planned and running DC & DR Site strategy and vice-versa. The proposed solution should have a capability to integrate with existing deployed cloud platform running Visualized and physical hardware. If required, bidder can leverage existing replication tool.	Same as query#16

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
72	Page no 21	Technical Specifications & Compliances	Solution should support Virtual Network visibility with application-centric protection from network threats and automation of common networking operations	Modification: Solution should support Virtual Network visibility with application-centric protection from network threats with automation of common existing network operations, security appliances & firewall, load balancer, backup solution. Justification: for seamless integration with existing network, load balancer and security devices. please include the ask modification..	Same as query#17
73	Page no 22	Technical Specifications & Compliances	The management tool should be able to provide global resource pooling and policy management to enable policy-based automation. The proposed management solution should provide proactive security & software advisory alerts and should outline the fixes required to address the issues and the management software should provide a mobile app for monitoring & viewing the resources being managed in the infrastructure	Modification: The management tool should be able to provide global resource pooling and policy management to enable policy-based automation. The proposed management solution should provide proactive security & software advisory alerts and should outline the fixes required to address the issues . Justification: "Security Breach" Please Remove the ask of Mobile app for monitoring & viewing the resources being managed in the infrastructure, as allowing Mobile app for monitoring & viewing the resources will be point of security breach in environment & will have security impact in SDC environment "	Same as query#18
74	Page no 22	Technical Specifications & Compliances	Switch should support minimum 1000 VRF instances with route leaking functionality, minimum 400K IPv4 LPM routes, min 16K Multicast routes, 3. minimum 90K no. of MAC addresses, minimum 32 MB Packet Buffer with intelligent shared-memory buffered architecture to simplify the system buffer management and queuing implementation.	Please amend the clause as: Switch should support minimum 256 VRF instances with route leaking functionality, minimum 120K IPv4 LPM routes, min 7K Multicast routes, 3. minimum 90K no. of MAC addresses, minimum 32 MB Packet Buffer with intelligent shared-memory buffered architecture to simplify the system buffer management and queuing implementation. Justification: 256 VRF are more than sufficient and higher VRF required in ISP environment which supports multiple tenant. 7K multicast routes more than sufficient for current deployment scenario, not more than 100 Multicast will be used.	Same as query#19
75	Page no 23	Technical Specifications & Compliances	Switch should support VXLAN to achieve Network Virtualisation, along with VXLAN and EVPN symmetric IRB. The switch should support BGP EVPN Route Type 2, Type 4 and Route Type 5 for the overlay control plane	Please amend the clause: Switch should support VXLAN to achieve Network Virtualisation, along with VXLAN and EVPN symmetric IRB. The switch should support BGP EVPN Route Type 2, Type 3/Type4 and Route Type 5 for the overlay control plane Justification: request to amend the clause for wider participation.	Same as query#20

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
76	Page no 23		All ports on Switch must provide encryption of traffic ie. MAC Sec Encryption (802.1AE) in hardware and should support QoS marking/priority settings of the end points as per the defined policy and should support Rate Limiting - Policing and/or shaping	Please amend the clause: All ports on Switch must provide encryption of traffic ie. MAC Sec Encryption (802.1AE) in hardware or should support Port security and should support QoS marking/priority settings of the end points as per the defined policy and should support Rate Limiting - Policing and/or shaping. Justification: MACsec is a point to point layer 1 encryption, in order to run Macsec across the campus, it's a prerequisite that MACsec is supported across all platforms (access to core). Additionally Macsec devices are more expensive as they require a hardware based encryption engine inbuild to carry out the line rate layer 1 encryption of traffic. Such requirements are an overkill within a campus environment, even in high security centric deployments, MACsec is only run over the links where the fibre is traversing/deployed in area which is outside the administration/control of the enterprise. Also	Same as query#21
77	Page no 23	Technical Specifications & Compliances	All ports on Switch must provide encryption of traffic ie. MAC Sec Encryption (802.1AE) in hardware and should support QoS marking/priority settings of the end points as per the defined policy and should support Rate Limiting - Policing and/or shaping	Please amend the clause: All ports on Switch must provide encryption of traffic ie. MAC Sec Encryption (802.1AE) in hardware or should support Port security and should support QoS marking/priority settings of the end points as per the defined policy and should support Rate Limiting - Policing and/or shaping Justification: MACsec is a point to point layer 1 encryption, in order to run Macsec across the campus, it's a prerequisite that MACsec is supported across all platforms (access to core). Additionally Macsec devices are more expensive as they require a hardware based encryption engine inbuild to carry out the line rate layer 1 encryption of traffic. Such requirements are an overkill within a campus environment, even in high security centric deployments, MACsec is only run over the links where the fibre is traversing/deployed in area which is outside the administration/control of the enterprise.	Same as query#21
78	Page no 14/Point 8	Eligibility/Qualification Criteria/Mandatory Undertakings	The Bidder shall not be blacklisted by any Central/State Government department or Corporation or Board or MP State PSU or Semi-Govt organisation as on date of bid submission.	Please amend the clause: The Bidder and OEM (Hardware and Software OEM's) shall not be blacklisted by any Central/State Government department or Corporation or Board or MP State PSU or Semi-Govt organisation as on date of bid submission.	No Change

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
79	Page no 16	Scope of work	Resiliency: Data resiliency with FTT-1/RF-2, with future support for FTT-2/RF-3.	Modification : Resiliency: Data resiliency with FTT-2/RF-3 to support SLA & availability of Critical SDC workload , Solution must be offered with 1Ms response time with asked IOPS of 100000 70:30 R:W- 16KB Block size Justification: Resiliency for Critical SDC workload, FTT2/RF3 will provide 2 Node failover resiliency , This will ensure availability required for SDC critical workload	Same as query#8
80	Page no 20	Technical Specifications & Compliances	Offered Solution shall be a next generation software defined platform which shall offer the functionality of Hyper-converged with independent scaling of both Storage and Compute without any downtime. The solution should provide enterprise-class storage services using latest x86 server infrastructures without dependence on a separate Storage Area Network Storage & associated components such as SAN Switches & HBAs. Offered HCI system should be a software-centric system for delivering compute, storage and networking resources in a tightly integrated system	Modification Request: Offered Solution shall be a next generation software defined platform which shall offer the functionality of Hyper-converged with independent scaling of both Storage and Compute without any downtime. The solution should provide enterprise-class storage services using latest x86 server infrastructures without dependence on a separate Storage Area Network Storage & associated components such as SAN Switches & HBAs. Offered HCI system should be a software-centric system for delivering compute, storage and networking resources in a tightly integrated HCI solution to deliver asked HCI services Justification: In HCI solution offering , different OEM have different architecture to deliver Virtual machine with Performance Integrated with compute, storage and networking resources .	Same as query#9
81	Page no 21		Resource requirement: HCI Solution should be provided with N+1 configuration having maximum 12 nodes Hybrid cluster, each node providing latest Intel processors/ AMD processors with base frequency of 2.1GHz or higher and DDR5 Memory. Solution must be proposed with at least 1500 or higher cores with minimum 1300 usable cores, 6 TB or higher memory and with minimum 500 TB usable storage without considering any deduplication, compression, erasure coding or any saving techniques benefits. Complete solution must be proposed with RF2/FTT1 i.e. two copies of data across the cluster. Each server/node should be configured with minimum of one Quad Port 10/25G network ports along having required 25G SFPs/DAC Cables for uplink connectivity	Resource requirement: HCI Solution should be provided with N+1 configuration having maximum 12 nodes Hybrid cluster, each node providing latest Intel processors/ AMD processors with base frequency of 2.1GHz or higher and DDR5 Memory. Solution must be proposed with at least 1500 or higher cores with minimum 1300 usable cores, 6 TB or higher memory and with minimum 500 TB usable storage without considering any deduplication, compression, erasure coding or any saving techniques benefits. Complete solution must be proposed with RF3/FTT2 i.e. two copies of data across the cluster. Each server/node should be configured with minimum of one Quad Port 10/25G network ports along having required 25G SFPs/DAC Cables for uplink connectivity Justification: Resiliency for Critical SDC workload & to ensure SLA with minimum 2 Node Failover architecture	Same as query#10

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
82	Page no 21	Technical Specifications & Compliances	The proposed HCI solution should be a factory shipped engineered & integrated appliance. The HCI solution should support SSD/NLSAS Drives. The HCI solution should support scalability up to 32 nodes in a single cluster. Each server node should have dedicated redundant hot swap power supply, cooling fans. There should not be any separate disaggregated components in the proposed HCI solution i.e. The 3 Major components of the HCI node i.e. Compute, RAM and Storage should be physically inside the node, no resource should be provisioned from outside the HCI Node.	Modification: The proposed HCI solution should be a factory shipped engineered & integrated appliance. The HCI solution should be offered with NVMe Drives only. The HCI solution should support scalability up to 16 nodes in a single cluster. Each server node should have dedicated redundant hot swap power supply, cooling fans. There should not be any separate disaggregated components in the proposed HCI solution i.e. The 3 Major components of the HCI node i.e. Compute, RAM and Storage should be physically inside the node, no resource should be provisioned from outside the HCI Node or as per OEM architecture ensuring full software defined architecture delivering required performance & IOPS. Justification: 1. Please Change asked Drives to NVMe Drives only , as NL SAS/ SSD /SAS have no Future road Map , Considering this is SDC Critical Environmnet any componnet which is going to be obsolute in near future should not be considered . 2. Change asked architecture with NVMe Drives only , This will ensure performance, abalability & high resileancy with Latest NVMe architctire. 2. Request Change in asked Cluster Scalability as 16/32 Nodes , since 32 Node in single cluster as SDS /HCI solution is not recommended from Management & for future Upgrade & Updated point of view.	Same as query#11
83	Page no 21	Technical Specifications & Compliances	The HCI solution shall support scaling Compute and Storage capacity and performance linearly by addition of nodes. VMs on existing nodes shall get the storage performance & capacity that gets enhanced by the addition of new HCI nodes in which HCI solution must provide on the fly change of ESE(Enterprise Storage Efficiency)-Deduplication/Compression for workload without any visible impact on storage and their operations and the bidder must ensure minimum 100,000 IOPS from the proposed HCI solution, assuming 8kb block size, and 70:30 read: write ratio.	Modification: The HCI solution shall support scaling Compute and Storage capacity and performance linearly by addition of nodes. VMs on existing nodes shall get the storage performance & capacity that gets enhanced by the addition of new HCI nodes in which HCI solution must provide on the fly change of ESE(Enterprise Storage Efficiency)-Deduplication/Compression for workload without any visible impact on storage and their operations and the bidder must ensure minimum 100,000 IOPS from the proposed HCI solution, assuming 8kb block size, and 70:30 read: write ratio ensuring less than 1ms response time . Justification: Since SDC application & DB Requirement is Critical from Performance POV. Pls define Response time to ensure Performance delivery & architecture is offered	Same as query#12

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
84	Page no 21	Technical Specifications & Compliances	The HCI software should pool all HDDs from all the nodes in the cluster to present a single storage resource pool to all server nodes in the cluster. The HCI storage should be a scale-out distributed storage. Proposed HCI solution must have a primary copy of the VM data locally to the Server/Node to offer faster read and write and secondary copy must be distributed on the other HCI Nodes. The intelligent software should consolidate the resources from each server node into a shared resource pool, delivering a high degree of flexibility while also simplifying management.	Modification: The HCI software should pool all NVMe HDDs from all the nodes in the cluster to present a single storage resource pool to all server nodes in the cluster. The HCI storage should be a scale-out distributed storage. Proposed HCI solution must have a primary copy of the VM data locally to the Server/Node to offer faster read and write and secondary copy must be distributed on the other HCI Nodes or as per OEM architecture ensuring less than 1ms sustainable response time. The intelligent software should consolidate the resources from each server node into a shared resource pool, delivering a high degree of flexibility while also simplifying management. Justification: Please change asked Disk type of Storage System to : NVMe. Kindly modify this clause for participation and better protection of investment.	Same as query#13
85	Page no 21	Technical Specifications & Compliances	The solution should automatically rebalance data to maintain balanced utilization of storage across the HCI nodes. When storage capacity is scaled up or scaled out, the HCI nodes must automatically redistribute data equally across all nodes equally without migrating VMs. Hypervisor software must provide Data at rest encryption with Native KMS which protects unauthorized data access.	Modification: The solution should automatically rebalance data to maintain balanced utilization of storage across the HCI nodes. When storage capacity is scaled up or scaled out, the HCI nodes must automatically redistribute data equally across all nodes equally without migrating VMs. Hypervisor / HCI solution must provide Data at rest encryption with Native KMS which protects unauthorized data access. Justification: Please change asked Hypervisor Based encryption to HCI Solution based Data at rest encryption - Option for Hardware based Encryption, as per OEM architecture to deliver Data at rest encryption feature	Same as query#14
86	Page no 21	Technical Specifications & Compliances	HCI solution should support Block, File and Object (S3) natively or using third party solution from Day1. Solution should support file storage supporting NFS v3/v4 and SMB 2.0/3.0 for Linux and Windows Guest with unlimited shares v with Active directory/LDAP	This is Restrictive Clauses, Please Remove this Clause / Modify : HCI solution should support Block, File natively or using third party solution from Day1. Solution should support file storage supporting NFS v3/v4/SMB 2.0/3.0 for Linux and Windows Guest with unlimited shares v with Active directory/LDAP Justification: This is Restrictive Clauses, This point is OEM specific to one OEM Request you to remove the ask of S3 protocol for wider participation of OEM	Same as query#15
87	Page no 21	Technical Specifications & Compliances	The solution quoted shall provide VM level/ Native Storage replication from a Primary site for planned DR Site strategy and vice-versa. If required, bidder can leverage existing replication tool.	The solution quoted shall provide VM level/ Native Storage replication from a Primary site for planned and running DC & DR Site strategy and vice-versa. The proposed solution should have a capability to integrate with existing deployed cloud platform running Visualized and physical hardware. If required, bidder can leverage existing replication tool.	Same as query#16

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
88	Page no 21	Technical Specifications & Compliances	Solution should support Virtual Network visibility with application-centric protection from network threats and automation of common networking operations	Modification: Solution should support Virtual Network visibility with application-centric protection from network threats with automation of common existing network operations, security appliances & firewall, load balancer, backup solution. Justification: for seamless integration with existing network, load balancer and security devices. please include the ask modification..	Same as query#17
89	Page no 22	Technical Specifications & Compliances	The management tool should be able to provide global resource pooling and policy management to enable policy-based automation. The proposed management solution should provide proactive security & software advisory alerts and should outline the fixes required to address the issues and the management software should provide a mobile app for monitoring & viewing the resources being managed in the infrastructure	Modification: The management tool should be able to provide global resource pooling and policy management to enable policy-based automation. The proposed management solution should provide proactive security & software advisory alerts and should outline the fixes required to address the issues . Justification: "Security Breach" Please Remove the ask of Mobile app for monitoring & viewing the resources being managed in the infrastructure, as allowing Mobile app for monitoring & viewing the resources will be point of security breach in environment & will have security impact in SDC environment "	Same as query#18
90	Page no 22	Technical Specifications & Compliances	Switch should support minimum 1000 VRF instances with route leaking functionality, minimum 400K IPv4 LPM routes, min 16K Multicast routes, 3. minimum 90K no. of MAC addresses, minimum 32 MB Packet Buffer with intelligent shared-memory buffered architecture to simplify the system buffer management and queuing implementation.	Please amend the clause as: Switch should support minimum 256 VRF instances with route leaking functionality, minimum 120K IPv4 LPM routes, min 7K Multicast routes, 3. minimum 90K no. of MAC addresses, minimum 32 MB Packet Buffer with intelligent shared-memory buffered architecture to simplify the system buffer management and queuing implementation. Justification: 256 VRF are more than sufficient and higher VRF required in ISP environment which supports multiple tenant. 7K multicast routes more than sufficient for current deployment scenario, not more than 100 Multicast will be used.	Same as query#19
91	Page no 23	Technical Specifications & Compliances	Switch should support VXLAN to achieve Network Virtualisation, along with VXLAN and EVPN symmetric IRB. The switch should support BGP EVPN Route Type 2, Type 4 and Route Type 5 for the overlay control plane	Please amend the clause: Switch should support VXLAN to achieve Network Virtualisation, along with VXLAN and EVPN symmetric IRB. The switch should support BGP EVPN Route Type 2, Type 3/Type4 and Route Type 5 for the overlay control plane Justification: request to amend the clause for wider participation.	Same as query#20

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
92	Page no 23		All ports on Switch must provide encryption of traffic ie. MAC Sec Encryption (802.1AE) in hardware and should support QoS marking/priority settings of the end points as per the defined policy and should support Rate Limiting - Policing and/or shaping	Please amend the clause: All ports on Switch must provide encryption of traffic ie. MAC Sec Encryption (802.1AE) in hardware or should support Port security and should support QoS marking/priority settings of the end points as per the defined policy and should support Rate Limiting - Policing and/or shaping. Justification: MACsec is a point to point layer 1 encryption, in order to run Macsec across the campus, it's a prerequisite that MACsec is supported across all platforms (access to core). Additionally Macsec devices are more expensive as they require a hardware based encryption engine inbuilt to carry out the line rate layer 1 encryption of traffic. Such requirements are an overkill within a campus environment, even in high security centric deployments, MACsec is only run over the links where the fibre is traversing/deployed in area which is outside the administration/control of the enterprise. Also	Same as query #21
93	Page no 23	Technical Specifications & Compliances	All ports on Switch must provide encryption of traffic ie. MAC Sec Encryption (802.1AE) in hardware and should support QoS marking/priority settings of the end points as per the defined policy and should support Rate Limiting - Policing and/or shaping	Please amend the clause: All ports on Switch must provide encryption of traffic ie. MAC Sec Encryption (802.1AE) in hardware or should support Port security and should support QoS marking/priority settings of the end points as per the defined policy and should support Rate Limiting - Policing and/or shaping Justification: MACsec is a point to point layer 1 encryption, in order to run Macsec across the campus, it's a prerequisite that MACsec is supported across all platforms (access to core). Additionally Macsec devices are more expensive as they require a hardware based encryption engine inbuilt to carry out the line rate layer 1 encryption of traffic. Such requirements are an overkill within a campus environment, even in high security centric deployments, MACsec is only run over the links where the fibre is traversing/deployed in area which is outside the administration/control of the enterprise.	Same as query #21

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
94	Page no. 13	2. Eligibility / Qualification Criteria Point # 3 Experience in Data Centre or Disaster Recovery site projects of IT infra implementation and O&M	The Bidder must have Successfully completed an assignment of similar nature Order involving Installation along with duly Configuring, and Maintenance of Data Centre or work. Disaster Recovery site projects of IT infra implementation and O&M within India. Provided that project shall not be of the period prior to 1st April 2020 and must have undertaken for any of the following: A) State or Central Government the order. OR B) State or Central PSU OR C) Scheduled Commercial Banks (as per RBI's latest notification)	As most of the IT Infra project has the tenure of 3 or 5 years, hence we understand that the date of completion of the Project tenure should NOT be Prior to 1st April 2020, - Kindly confirm.	Refer Corrigendum-2 Sn#12
95	Page no. 24	Project Timelines & Payment Milestones Point 1	Supply, Installation and Commissioning 1) Supply of complete Infrastructure Timeline : T+6 weeks	Normally the Procurement & delivery of the IT infrastructure from OEM takes at least 8 to 10 weeks, hence 6 weeks would be too short and would lead us to undue penalty. We hereby request Authority to kindly amend the timeline as T+10 weeks	Refer Corrigendum-2 Sn#1, 2 & 3
96	Page no. 24	5. Project Timelines & Payment Milestones Point 1, 2 & 3	Payable : 1) 20% of the Work Order Value 2) 20% of the Work Order Value 3) 50% of the Work Order Value	As per the standard payment practice followed in the similar tenders for IT Infrastructure projects, the payment terms are : 1) 70% of the Work Order Value on Supply & Delivery 2) 20% against the Installation, Commissioning, FAT & Go-live We hereby request Authority to kindly amend the payment terms.	Refer Corrigendum-2 Sn#1, 2 & 3
97	page no. 18	Scope of Work 3.14 Inclusive Scope	connectivity	We understand that the bandwidth / connectivity would be provided by MPSDC. Kindly confirm. If bidder is responsible for providing the connectivity, then kindly specify the minimum bandwidth / connectivity requirement.	No Change - Internet bandwidth shall be provided by MPSDC, whereas SI shall be responsible for the connectivity of the installed system (IT and Non-IT) with the SDC electrical supply/panel. SI shall procure the required electrical components to establish the seamless connectivity.

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
98	07-Aug	EMD Exemption	EMD Exemption for Micro and Small Enterprises and Start-ups of Madhya Pradesh Micro and Small Enterprises (MSEs) registered in Madhya Pradesh are eligible for exemption from submitting the Earnest Money Deposit (EMD) The MSEs must be registered under the same or a similar service category as specified in the RFP. The bidder must ensure their MSEs registration aligns with the services outlined in the tender. Verification will be based on the classification stated in the Udyam Certificate. The exemption applies exclusively to the EMD requirement of the tender. All other terms and conditions specified in the RFP will remain applicable to MSEs.	We respectfully request you to kindly amend the clause to: EMD Exemption for Micro and Small Enterprises and Start-ups of Madhya Pradesh Micro and Small Enterprises (MSEs) registered in Madhya Pradesh other States/UTs across India also are eligible for exemption from submitting the Earnest Money Deposit (EMD) The MSEs must be registered under the same or a similar service category as specified in the RFP. The bidder must ensure their MSEs registration aligns with the services outlined in the tender. Verification will be based on the classification stated in the Udyam Certificate. The exemption applies exclusively to the EMD requirement of the tender. All other terms and conditions specified in the RFP will remain applicable to MSEs.	No Change
99	7	Bid Data Sheet	Rs. 54,00,000/- (Fifty-Four lakh only) to be paid online through e-procurement portal. EMD in the form of Bank guarantee shall be acceptable.	The current EMD amount is on the higher side and may restrict wider participation. Reducing the EMD to 1% will encourage healthy competition and enable participation from more eligible bidders.	Refer Corrigendum-2 Section: Bid Data Sheet Sn#1

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
100	13	3.Experience in Data Centre or Disaster Recovery site projects of IT infra implementation and O&M	The bidder must have successfully completed an assignment of similar nature involving the Supply, Installation, Configuring, and Maintenance of Data Centre or Disaster Recovery site projects of IT infra implementation and O&M within India Provided that project shall not be of the period prior to 1st April 2020 and must have undertaken for any of the following: A)State or Central Government OR B)State or Central PSU OR C)Scheduled Commercial Banks (as per RBI's latest notification) The below mentioned order values are applicable for the above criteria: The bidder should have undertaken One project of a similar nature with a minimum value of INR 14 Crores. OR Two projects of a similar nature, each with a minimum value of INR 10 Crores. OR Three projects of a similar nature, each with a minimum value of INR 07 Crores. Note: Only supply orders shall not be considered. The Bidder should have provided FMS/Data Centre Operation for such projects.	The bidder should have undertaken at least one project of similar nature involving DC and DR implementation with a minimum value of INR 11.33 Crores, executed within India on or after 1st April 2020, for any of the eligible organizations as defined in the RFP.	No Change
101	Page 21	HCI Tech specs point 10,	The solution quoted shall provide VM level / Native Storage replication from a Primary site for planned DR Site strategy.If requiered, bidder can leverage existing replication tool.	As mentioned in SOW, we understand that bidder needs to factor 3rd party tool for the requested functionality. We request MPSEDC to provide the required inputs to factor the 3rd party tool.	No Change MPSeDC will provide Carbonite tool for DR Replication, in case bidder wants to use any other tool for DR Replication, they must provision for the same.
102	16	Scope of work - Appliances	Supply factory Integrated appliances	Requests relaxation on the requirement for factory-integrated HCI appliances. A customized, OEM-certified HCI solution assembled using validated components and supported by the OEM shall also be considered compliant, provided it meets all functional, performance, and support requirements."	Refer Corrigendum-2 Sn#7
103	16	Scope of Work - 3.2	Hardware, Compute & Storage --Performance: Deliver a minimum of 100,000 IOPS in proposed HCI solution with software defined storage.	Need clarification of required IOPS are as per Cluster or Node or Disc?	No Change IOPS per cluster

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
104	16	Scope of Work - 3.3	Virtualization Platform & Integration - Integration: There should be seamless communication, VM migration with MPSEDC's existing virtualization environment and proposed HCI solution.	Need more clarification on existing Virtualization platform for designing the solution.	No Change Existing deployed virtualization - VMWare and Openstack
105	17	Scope of Work -3.7	3.7-Backup, Replication & Disaster Recovery : •Replication: Replication from HCI to Traditional Bare Metal and vice-versa	Need more clarification on existing Virtualization platform to understand the type of Replication tool we can explore.	No Change Existing deployed virtualization - VMWare and Openstack
106	20	3.16Technical Specification and Compliance	The proposed HCI solution should be a factory shipped engineered & integrated appliance. The HCI solution should support SSD/NLSAS Drives. The HCI solution should support scalability up to 32 nodes in a single cluster. Each server node should have dedicated redundant hot swap power supply, cooling fans. There should not be any separate disaggregated components in the proposed HCI solution i.e. The 3 Major components of the HCI node i.e. Compute, RAM and Storage should be physically inside the node, no resource should be provisioned from outside the HCI Node.	A customized, OEM-certified HCI solution assembled using validated components and supported by the OEM shall also be considered compliant, provided it meets all functional, performance, and support requirements."	Refer Corrigendum-2 Sn#7
107	21	Technical Specification and Compliance -9	HCI solution should support Block, File and Object (S3) natively or using third party solution from Day1. Solution should support file storage supporting NFS v3/v4 and SMB 2.0/3.0 for Linux and Windows Guest with unlimited shares integrated with Active directory/LDAP	Needs to understand Object Storage requirement use case	Will be discussed with successful bidder. Refer Corrigendum-2 Sn#17
108	22	Management - 5	The solution should support Single click non-disruptive rolling upgrades of HCI software and system firmware's. The solution should Track, report, and view trends for metrics like CPU, memory, IOPs, latency etc. and solution should provide insight into the performance, configurations and capacity problems. It should also associate workflows with alerts to automatically initiate corrective measures at critical thresholds.	Upgrades for hypervisor and hardware firmware are handled through separate tools. There is no unified one-click lifecycle management integrating BIOS, firmware, drivers, and hypervisor updates.	No Change
109	23	OEM Turnover	The OEMs of the quoted products shall have a minimum average annual turnover of Rs. 500 Crores in each of the last three financial years, i.e., FY 2022-23, FY 2023-24, and FY 2024-25	Kindly Amend the clause either as per below: "The OEMs of the quoted products shall have a minimum average annual turnover of Rs. 200 Crores, calculated as an average over the last three financial years, i.e., FY 2022-23, FY 2023-24, and FY 2024-25." OR "The OEMs of the quoted products shall have a minimum Consolidated turnover of Rs. 500 Crores, of last three financial years, i.e., FY 2022-23, FY 2023-24, and FY 2024-25."	No Change

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
110	24	Experience in Data Centre or Disaster Recovery site projects of IT infra implementation and O&M	The bidder must have successfully completed assignment of similar nature involving Installation, Configuring, and Maintenance of Data Centre or Disaster Recovery site projects of IT infra implementation and O&M within India Provided that project shall not be of the period prior to 1st April 2020 and must have undertaken for any of the following: A) State or Central Government OR B) State or Central PSU OR C) Scheduled Commercial Banks (as per RBI's latest notification) The below mentioned order values are applicable for the above criteria the bidder should have undertaken One project of a similar nature with a minimum value of INR 14 Crores. OR Two projects of a similar nature, each with a minimum value of INR 10 Crores. OR Three projects of a similar nature, each with a minimum value of INR 07 Crores. Note: Only supply orders shall not be considered. The Bidder should have provided FMS/Data Centre Operation for such projects.	Kindly amend the eligibility clause pertaining to the completed assignment(s) of similar nature involving installation, configuration, and maintenance of Data Centre or Disaster Recovery (DR) site projects, including IT infrastructure implementation and supply of servers and storage within India, as follows: The bidder should have successfully completed: One (1) project of a similar nature with a minimum value of INR 04 Crores, OR Two (2) projects of a similar nature, each with a minimum value of INR 02 Crores, OR Three (3) projects of a similar nature, each with a minimum value of INR 1.5 Crores.	No Change
111	12	Eligibility/Qualification Criteria Point No 2	Financial Turnover - Average Annual Turnover of the Bidder shall be atleast Rs. 50 Crore in India during last three years	As per General Financial Rules (GFR) 2017 and the Manual for Procurement of Goods and Public Procurement Policy for Micro and Small Enterprises (MSEs) Order, 2012, issued under Section 11 of the MSMED Act, 2006, provides for special procurement benefits to MSEs, including relaxation in eligibility conditions to enhance participation of MSEs in Government tenders. We request for exemption of this criteria for MSE Bidders	No Change Refer EMD Exemption of Bid Data Sheet Section

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
112	13-14	Eligibility/Qualification Criteria Point No 3	Experience in Data Center or Disaster Recovery site projects of IT Infra Implementation and O&M After April 2020 onwards, State gov/ Central/ Banks-Scheduled Commericals - One project 14 Cr. - Two project 10 Cr. - Three projects 7 Cr. with completion. Note: only supply orders not considered, FMS/ Data centre operation for such projects required.	As per General Financial Rules (GFR) 2017 and the Manual for Procurement of Goods and Public Procurement Policy for Micro and Small Enterprises (MSEs) Order, 2012, issued under Section 11 of the MSMED Act, 2006, provides for special procurement benefits to MSEs, including relaxation in eligibility conditions to enhance participation of MSEs in Government tenders. We request for exemption of this criteria for MSE Bidders	No Change Refer EMD Exemption of Bid Data Sheet Section
113	12	Eligibility/Qualification Criteria Point No 1	Legal Entity, since five years. No JV/Consortium is allowed.	Request to Allow JV/Consortium	No Change
114	20	4. Technical Specification and Compliance	The proposed HCI Solution should be a factory shipped engineered & integrated appliance. The HCI Solution should support SSD/NLSAS Drives. The HCI solution should support scalability up to 32 nodes in single cluster. Each server node should have dedicated redundant hot swap power supply, cooling fans. There should not be any separate disaggregated components in the proposed HCI solution i.e. The 3 Major components of the HCI node i.e. Compute, RAM and storage should be physically inside the node, no resource should be provisioned from outside the HCI Node.	We understand that as per the RFP, the requirement is for an HCI solution. However, you have mentioned a factory-shipped, engineered, and integrated appliance solution, which results in vendor lock-in and restricts the solution to a single vendor only. Due to this requirement, we are unable to comply using our world-class VMware vSAN-based HCI solution. We therefore request you to kindly remove the highlighted and underlined paragraph.	Refer Corrigendum-2 Sn#7
115	21	4. Technical Specification and Compliance	Proposed Solution must support at least two industry standard hypervisors like ESXi, AHV, Hyper-V and also allow migration from above mentioned Hypervisor. Bidders need to submit the public URL document on the same.	This requirement is again specific to a single vendor. We request that it should be removed to ensure a fair opportunity and broader participation from the market. Or it should be modified as per following . Proposed Solution must support at least one industry leader hypervisors like vSphere ESXi, Hyper-V, AHV/KVM and also allow migration from existing running workload from any Hypervisor vSphere ESXi, Hyper V and AHV/kVM. Bidders need to submit the public URL document on the same.	Refer Corrigendum-2 Sn#18

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
116	21	4. Technical Specification and Compliance	The HCI software should pool all HDDs from all the nodes in the cluster to present a single storage resource pool to all server nodes in the cluster. The HCI storage should be a scale-out distributed storage. Proposed HCI solution must have a primary copy of the VM data locally to the Server/Node to offer faster read and write and secondary copy must be distributed on the other HCI Nodes. The intelligent software should consolidate the resources from each server node into a shared resource pool, delivering a high degree of flexibility while also simplifying management.	Every HCI OEM has different algorithm to manage VM data copy either locally or distributed manner. Hence we request to modify this clause with Local or distributed copy for VM data. This change will provide a fair chance for our participation in this bid. The HCI software should pool all HDDs from all the nodes in the cluster to present a single storage resource pool to all server nodes in the cluster. The HCI storage should be a scale-out distributed storage. Proposed HCI solution must have a primary copy of the VM data distributed / locally to the Server/Node to offer faster read and write and secondary copy must be distributed on the other HCI Nodes. The intelligent software should consolidate the resources from each server node into a shared resource pool, delivering a high degree of flexibility while also simplifying management.	Refer Corrigendum-2 Sn#19
117	21	4. Technical Specification and Compliance	HCI solution should support Block, File and Object (S3) natively or using third party solution from Day1. Solution should support file storage supporting NFS v3/v4 and SMB 2.0/3.0 for Linux and Windows Guest with unlimited shares integrated with Active directory/LDAP	We support 100 to 500 shares for file service with NFS and SMB. Hence, requesting to modify this clause instead of unlimited shares which provide benefit to single vendor only. Therefore, this clause should be modified as per following . HCI solution should support Block, File and Object (S3) natively or using third party solution from Day1. Solution should support file storage supporting NFS v3/v4 and SMB 2.0/3.0 for Linux and Windows Guest shares integrated with Active directory/LDAP	Refer Corrigendum-2 Sn#17
118	22	4. Technical Specification and Compliance	The solution should enable integration of third-party network and security solutions through open architecture and standard APIs. The bidder shall provide a list of ecosystem vendors that integrate with the framework and should support micro-segmentation for Bare Metal & Virtual end points by using VM attributes like VM name, OS, MAC address, IP etc. Solution should also support for L4-L7 integration with policy-based redirect along with any OEM firewall.	VMware HCI solution support Microsegmentation with NSX and third party OEMs solution for virtual machines only. Hence we request to modify this clause as per following. The solution should enable integration of third-party network and security solutions through open architecture and standard APIs. The bidder shall provide a list of ecosystem vendors that integrate with the framework and should support micro-segmentation for Bare Metal / Virtual end points by using VM attributes like VM name, OS, MAC address, IP etc. Solution should also support for L4-L7 integration with policy-based redirect along with any OEM firewall.	No Change

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
119	22	4. Technical Specification and Compliance	All proposed hardware and software must be supported 24X7X365 Technical support for 5 yrs. with NBD RMA and must be enabled from day one.	We think NBD RMA means Next business Day and Remote Mode Access. We suppose MPSEDC will be using HCI Solution for business critical application where uptime would be top most priority and in case any issue arise the response time from the respective OEM should be 30 minutes. Hence, this clause should be modified as per following . All proposed hardware and software must be supported 24X7X365 Technical support for 5 yrs. with 30 minutes response time and unlimited incidents support.	Refer Corrigendum-2 Sn#20
120	22	4. Technical Specification and Compliance	Addition Point	We understand as per RFP that the ask is for an Leading Enterprise Virtualization technology with HCI solution, we see that there are no technical asks mentioned for this enterprise grade virtualization solution. We suggest that the below minimum specs for a enterprise grade virtualization solution be included in the RFP 1. Solution should include compute Virtualization layer that sits directly on the bare metal server hardware with no dependence on a general purpose OS with features like proactive HA, DRS, replication, fault tolerance with continuous availability of VMs with zero downtime and zero data loss, hot add of CPU, memory, devices for windows as well as Linux VMs, VM level encryption, secure boot, uninterrupted service delivery within and across datacenter at geographical distance (<100ms latency), distributed virtual switch, kernel embedded network and storage virtualization technology. 2. Live migration of VM disk from one storage array to another without any VM downtime. Support this migration from one storage protocol to another e.g.: FC, NFS, iSCSI, DAS. 3. Proactive High availability capability that utilizes server health information and migrates VMs from degraded hosts before problem occurs. 4. Should support High Availability for migration of VMs in case	No Change

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
121			As per Eligibility Criteria Sr.No 3 - The bidder must have successfully completed an assignment of similar nature involving the Supply, Installation, Configuring, and Maintenance of Data Centre or Disaster Recovery site projects of IT infra implementation and O&M within India	The bidder must have successfully completed an assignment of similar nature involving the Supply, Installation, Configuring, and Maintenance of Data Centre or Disaster Recovery site projects of Data Centre or Disaster Recovery site projects of IT infra implementation and O&M/Warranty Support within India	No Change
122			Project Timelines and Payment Milestones :Sr.No 1 Supply of Complete Infrastructure as per Financial Proposal	70 % of the Wok Order Value	Refer Corrigendum-2 Sn#1, 2 & 3
123			Project Timelines and Payment Milestones :Sr.No 2 Installation, Commissioning of all infrastructure and system UAT/ Verification	20% of the Work Order Value	Refer Corrigendum-2 Sn#1, 2 & 3
124			Project Timelines and Payment Milestones :Sr.No 3 Integration with Existing Infra/Solution and FAT of all the Installed Commisioned Infrastructure and Go-Live (System is in Usable State)		Refer Corrigendum-2 Sn#1, 2 & 3
125			SLAs for Supply, Installation and Commissioning of Infrastructure Sr.No 1- Supply of Complete Infrastructure (as per BOQ)	T + 10 Weeks	Refer Corrigendum-2 Sn#1, 2 & 3
126			As per Scope of work, 3.12- Licensing & Update- Upgrade : If any upgrade to the solution / Infra is required to smoothly run the system, the Bidder/ OEM shall update / upgrade / Replace at no separate cost to MPSEDC within the complete AMC period.	The bidder's scope of responsibility shall be limited to the supply, installation, Update and upgrade of the quoted hardware and associated software licenses. Any compatibility issues arising from the existing infrastructure, legacy applications, or third party software already deployed in the client environment shall not fall under the bidder's scope. The bidder will ensure that the supplied hardware and licensed software function as per OEM specifications.	Refer Corrigendum-2 Sn#21
127	16	Scope of work - Solution design & architecture	•Cluster Configuration: Maximum 12-node cluster scalable to 32 nodes.	The rationale for prescribing a fixed 12-node cluster architecture is unclear. Since the RFP already specifies an initial requirement of 1,500 cores with minimum 1,300 usable cores, it is requested that scalability be defined in terms of overall compute capacity rather than node count. Kindly revise the clause to state that the solution should support present requirements of ~1,300 usable cores and be scalable in future to 2,000–2,500 cores or higher, irrespective of underlying cluster size or node architecture.	Refer Corrigendum-2 Sn#7

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
128	16	Scope of work - Solution design & architecture	Data resiliency with FTT- 1/RF-2, with future support for FTT-2/RF-3.	The terms FTT/RF are OEM-specific constructs. It is requested to replace these with outcome-based requirements such as defined SLA for availability, durability, and data loss (RPO/RTO), allowing bidders to propose architectures that meet or exceed these service levels using their native technologies.	Refer Corrigendum-2 Sn#5
129	16	Scope of work - Virtualization platform & Integration	There should be seamless communication, VM migration with MPSEDC's existing virtualization environment and proposed HCI solution	Seamless VM migration across heterogeneous HCI platforms is not technically feasible unless both environments belong to the same OEM stack. Request to remove the VM migration requirement and revise the clause as follows: "There should be seamless communication between MPSEDC's existing virtualization environment and the proposed HCI solution."	No Change
130	20	List of Items	N+1 configuration with Maximum 12-node cluster scalable upto 32 nodes	Similar to the earlier cluster clause, prescribing node counts unnecessarily restricts solution architectures. Kindly define requirements in terms of N+1 resiliency and overall compute scalability, with present usable capacity of ~1,300 cores and future expansion to 2,000–2,500 cores or higher, independent of node quantity.	Refer Corrigendum-2 Sn#7
131	20	Technical Specifications & Compliances	The proposed HCI solution should be a factory shipped engineered & integrated appliance. The HCI solution should support SSD/NLSAS Drives. The HCI solution should support scalability up to 32 nodes in a single cluster. Each server node should have dedicated redundant hot swap power supply, cooling fans. There should not be any separate disaggregated components in the proposed HCI solution i.e. The 3 Major components of the HCI node i.e. Compute, RAM and Storage should be physically inside the node, no resource should be provisioned from outside the HCI Node.	Request to remove the mandatory 32-node scalability condition. Proposed revision: "The proposed HCI solution should be a factory-shipped engineered and integrated appliance supporting SSD/NLSAS drives. Each server node should have redundant hot-swap power supplies and cooling fans. There should be no disaggregated components; Compute, RAM, and Storage must reside within the node, with no external resource dependency."	Refer Corrigendum-2 Sn#7
132	20	Technical Specifications & Compliances	Proposed Solution must support at least two industry standard hypervisors like ESXi, AHV, Hyper-V and also allow migration from above mentioned Hypervisor. Bidders need to submit the public URL document on the same	The requirement for specific third-party hypervisors limits participation of engineered systems offering native virtualization. Request removal of ESXi/AHV/Hyper-V mandate and migration dependency. Bidders should be allowed to propose native virtualization platforms, provided functional equivalence and workload portability are ensured.	Refer Corrigendum-2 Sn#18

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133	20	Technical Specifications & Compliances	HCI Solution should be provided with N+1 configuration having maximum 12 nodes Hybrid cluster, each node providing latest Intel processors/ AMD processors with base frequency of 2.1GHz or higher and DDR5 Memory. Solution must be proposed with at least 1500 or higher cores with minimum 1300 usable cores, 6 TB or higher memory and with minimum 500 TB usable storage without considering any deduplication, compression, erasure coding or any saving techniques benefits. Complete solution must be proposed with RF2/FTT1 i.e. two copies of data across the cluster. Each server/node should be configured with minimum of one Quad Port 10/25G network ports along having required 25G - SFPs/DAC Cables for uplink connectivity	Request to remove references to maximum node count and RF/FTT constructs. Proposed clause: "HCI Solution should offer Intel/AMD processors with base frequency ≥ 2.1 GHz and DDR5 memory. The solution must provide minimum 1,500 cores with at least 1,300 usable cores, 6 TB or higher memory, and minimum 500 TB usable storage (without deduplication/compression benefits). Each server/node should have at least one Quad-Port 10/25G NIC with required 25G SFPs/DAC cables for uplink connectivity	Refer Corrigendum-2 Sn#7
134	21	Technical Specifications & Compliances	Must provide on the fly change of ESE(Enterprise Storage Efficiency)- Deduplication/Compression for workload without any visible impact on storage and their operations and the bidder must ensure minimum 100,000 IOPS from the proposed HCI solution, assuming 8kb block size, and 70:30 read: write ratio.	On-the-fly deduplication/compression is OEM-specific. Instead, it is suggested to specify storage performance and efficiency outcomes (IOPS, latency, usable capacity) and allow bidders to meet these using native storage architectures or alternative storage technologies.	No Change
135	21	Technical Specifications & Compliances	The HCI software should pool all HDDs from all the nodes in the cluster to present a single storage resource pool to all server nodes in the cluster. The HCI storage should be a scale-out distributed storage. Proposed HCI solution must have a primary copy of the VM data locally to the Server/Node to offer faster read and write and secondary copy must be distributed on the other HCI Nodes. The intelligent software should consolidate the resources from each server node into a shared resource pool, delivering a high degree of flexibility while also simplifying management.	This clause mandates a specific storage architecture. Since different OEMs implement distributed storage differently, request removal of this clause and replacement with performance, availability, and scalability objectives, ensuring architectural neutrality	Refer Corrigendum-2 Sn#19
136	21	Technical Specifications & Compliances	The solution should automatically rebalance data to maintain balanced utilization of storage across the HCI nodes. When storage capacity is scaled up or scaled out, the HCI nodes must automatically redistribute data equally across all nodes equally without migrating VMs. Hypervisor software must provide Data at rest encryption with Native KMS which protects unauthorized data access.	These are implementation-specific features. Request removal and replacement with outcome-based requirements covering data balancing, encryption at rest, and key management, allowing OEMs to meet these through their native platforms.	Refer Corrigendum-2 Sn#8

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137	23	Technical Specifications & Compliances	Network Switch (TOR):	TOR switch specifications appear aligned to a particular HCI implementation. As networking is internal to engineered systems, request removal of detailed TOR specifications. Instead, RFP may state that the proposed solution must include sufficient internal networking capacity to support the complete solution performance and scalability requirements.	Refer corrigendum
138	NA	Technical Specifications & Compliances	Please add these clauses for the betterment of the solution.	The solution should provide cloud-consistent APIs and management tools (CLI/SDK/IaC) equivalent to the bidder's public cloud to enable uniform provisioning and operations across on-prem and cloud environments. The platform should support seamless portability of workloads between on-premises and public cloud, including VM images and infrastructure configurations, to enable hybrid deployment and avoid vendor lock-in. The solution should be fully managed by the OEM across hardware, firmware, hypervisor, networking, and storage, including monitoring, patching, and lifecycle management. The platform should provide operator access control with customer visibility, approval-based access, session monitoring, and the ability to terminate OEM access. Loss of connectivity to the management/control plane should not impact running workloads or applications on the on-premises system. The solution should provide managed Kubernetes capability on-premises equivalent to the bidder's public cloud Kubernetes service. The platform should support optional GPU expansion for AI/ML and accelerated workloads within the same system	No Change

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
139	17	3.8 Data & Workload Migration by SI/bidder	Strategy: Prepare and execute a phased migration strategy including testing, validation, and rollback to ensure secure and minimal-downtime migration.	Please confirm our understanding: • Application owner will migrate, deploy the application. Application deployment, monitoring, modification, migration, code changes or tuning, management & troubleshooting is out of scope for bidder. Infra level support will be provided by bidder. • Resolution related to underlying infra issues & support for the application issues will be provided by the bidder.	Understanding is correct.
140	16	3.3 Virtualization Platform & Integration	Integration: There should be seamless communication, VM migration with MPSEDC's existing virtualization environment and proposed HCI solution.	Please help us understand the existing infra for seamless integration: • What is the current virtualization platform • How many VMs are currently hosted • How many applications are currently running, what is their code language, version, version	Current virtualization platforms are VMWare and open stack. However, VMs and other details will be shared with successful bidder.
141	17	3.7 Backup, Replication & Disaster Recovery	• Features: Support VM-level snapshots, cloning, native replication, and incremental synchronization using inbuilt features/3rd party tools provided with the solution. • DR Strategy: Bi-directional DC-DR-NDR replication and DR drills using inbuilt features/3rd party tools provided with the solution. • Replication: Replication from HCI to Traditional Bare Metal and vice-versa	Kindly share your answers in-line: Backup: • The needed backup policies • Backup retention period • Approximate backup size. DR: • DR setup will be for Production environment only? • How many DR Drill needs to be conducted in a year under normal conditions. • What will be the duration in hours for each DR drill? • What percentage needs to be assumed for DR site of DC (example: DR will be 50% of DC capacity) or same sizing needs to be considered for DR site? • What is the RTO & RPO requirement. • Please confirm DC & DR locations.	DC location is Bhopal, DR location - NDC Bhuwaneswar. DR setup will be for production environment. However, VMs and other details will be shared with successful bidder.
142	17	3.10 Operations & Support by SI/bidder	• Support Model: 24x7x365 Premium Support with proactive monitoring and incident management backed by OEM support. • Hardware: Next Business Day (NBD) hardware replacement and onsite support as required backed by OEM support.	Please confirm our understanding: • This support is needed for Hardware & Virtualization layer only or end-to-end support is needed like monitoring, on-call, L3 level support for virtual machines as well? • Managed Services will cover 24*7 standard support for P1 & P2 incidents & business hour support for P3 & P4 incidents. • Please confirm the premium support can be offline or any onsite resources are needed. • Please confirm the onsite location for these resources, is it Bhopal, Madhya Pradesh.	• Support should be provided for all compomnents by OEM/ SI • As per RFP SLA terms • Online as well as Offilne as required • DC is at Bhopal MP

#	RFP Page#	Clause/ Heading	Clause Description	QUERIES	RESPONSE
143	17	3.9 Training & Knowledge Transfer by SI/bidder	• Deliverables: Administrator and operational training, documentation, System Manual and Standard Operating Procedures (SOPs). • Sign-off: Obtain formal knowledge transfer sign-off from MPSEDC.	Kindly elaborate on each point: • To how many resources the training needs to be provided and for how many days, • Any specific number on the batch size, • Please confirm the training location • Number of times the training sessions needs to be conducted. • What materials (training manuals) & topics are expected here.	Will be shared with successful bidder.
144	All	All	Generic	Are there dependencies between applications or services that need to be replicated in the DR setup?	DR is already setup, but the proposed solution must ensure DR replication by Bidder/ OEM
145	18	3.14 Inclusive Scope	Throughput and Bandwidth should be as per the solution. The SI should ensure that there should not be any bottleneck pertaining to throughput and bandwidth, while transferring data end to end.	Kindly confirm the daily delta change of data in a day time (across 24 hrs) or the peak delta change at single point of time in a day, whichever is known.	Will be shared with successful bidder.
146	24	6. Service Level Agreement (SLA) & Penalties	Generic	What are required uptime SLA parameters?	As per RFP requirement
147	17	3.12 Licensing & Updates	Lifecycle: Provide all required licenses, updates, patches, and firmware upgrades for the entire contract duration.	Kindly elaborate lifecycle management is related to Hardware & Virtualization platform only or for the hosted VMs also. If for VMs also then kindly share each cluster wise sizing (CPU/Memory/Disk) details for- DB, Linux and Windows servers that should be in separate cluster for license.	Understanding is correct its related to supplied hardware and all solutions.
148	16	3. Scope of Work	The selected Bidder shall be responsible for the end-to-end planning, design, supply, installation, integration, testing, commissioning, operations support, and maintenance of an enterprise-grade on-premises Hyper-Converged Infrastructure (HCI) Solution at MPSEDC.	The expected solution needs to be completely on-prem or a solution that is hosted on-prem and the management plane in public-cloud is also a feasible option?	Completely on-prem Solution.
149	24	5. Project Timelines & Payment Milestones	Supply of complete Infrastructure as per Financial Proposal: T+ 6 weeks	We request you to extend the timelines by 2 more weeks i.e., T+8 weeks for "Supply of complete Infrastructure as per Financial Proposal" since it will not be just resell and expansion for the Sis whole new set of hardware needs to be procured and then installation setup and migration needs to be done.	Refer Corrigendum-2 Sn#1, 2 & 3
150	51	Annexure-8.2 - Financial Bid Form:	MS Cost for 5 years	There is no line item to embedd MS cost in the financial bid form, kindly suggest where to incorporate this cost.	Proposed solution should be inclusive of MS cost.
151	24	Sec 5	Installation + UAT timeline T+8 weeks	Request to revise Installation + UAT timeline from T+8 weeks to T+10 weeks considering migration, integration and testing complexity.	Refer Corrigendum-2 Sn#1, 2 & 3
152	24	Sec 5	FAT & Go-Live timeline T+10 weeks	Request to revise FAT & Go-Live timeline from T+10 weeks to T+12 weeks to ensure stable acceptance and cutover.	Refer Corrigendum-2 Sn#1, 2 & 3

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153	36	Sec 16	Liquidated Damages @1% per week	Request to revise LD from 1% per week of undelivered portion to 0.5% per week.	No Change
154	36	Sec 16	LD cap 10% of total contract value	Request to revise maximum LD cap from 10% to 5% of total contract value.	No Change
155	25	Sec 6.2	P1 SLA Response ≤15 mins, Resolution ≤4 hrs	Request to revise P1 Resolution to <8 hours with penalty of 0.01% to align with industry practice.	No Change
156	26	Sec 6.2	P2 SLA Response ≤30 mins, Resolution ≤8 hrs	Request to revise P2 Resolution to <12 hours with penalty of 0.01%.	No Change
157	26	Sec 6.2	P3 SLA Response ≤2 hrs, Resolution ≤24 hrs	Request to revise P3 penalty to a flat 0.01% considering minor impact nature.	No Change
158	18-19	Sec 3.14	Any service required deemed included	Request detailed clarification and boundary definition of activities covered under 'Any service required' to avoid scope ambiguity.	No Change
159	17	Section 3.7	Bi-directional DC-DR-NDR Replication	The RFP mandates bi-directional DC-DR-NDR replication, however DR/NDR hardware and related components are not part of the BOQ; please clarify whether bidders are required to include additional DR/NDR hardware and licenses or whether replication is expected to be implemented using existing customer DR/NDR infrastructure. If existing DR/NDR Infra is to be used please share the details of the corresponding	Replication is expected to be implemented using existing MPSeDC DR/NDR infrastructure. Bidder has to provision required replication tools.

Signature Not Verified

Digitally signed by MANOJ KUMAR SAXENA
Date: 2026.02.16 13:04:31 IST
Location: Madhya Pradesh-MP

